

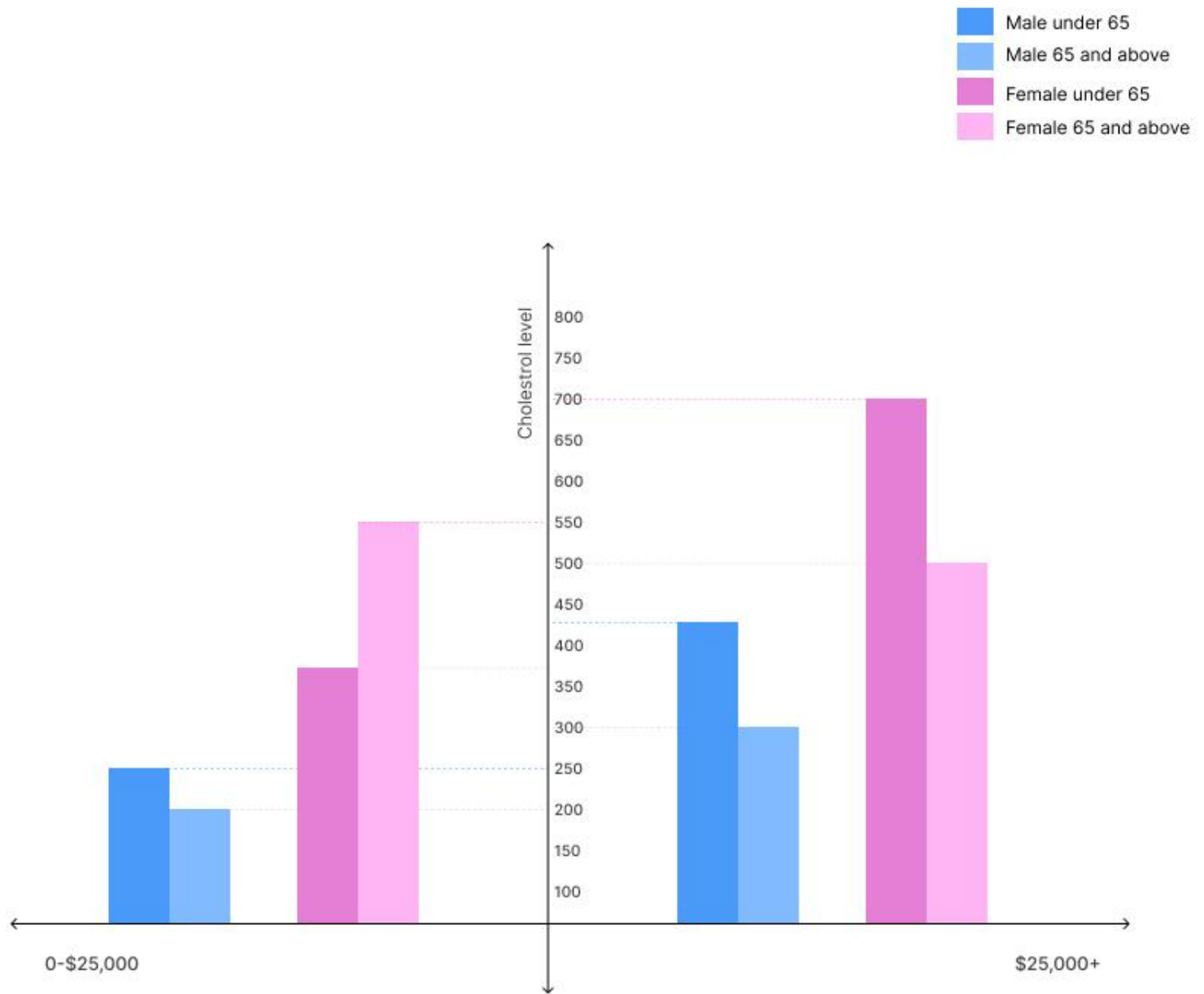
01. Introduction to Data Visualization

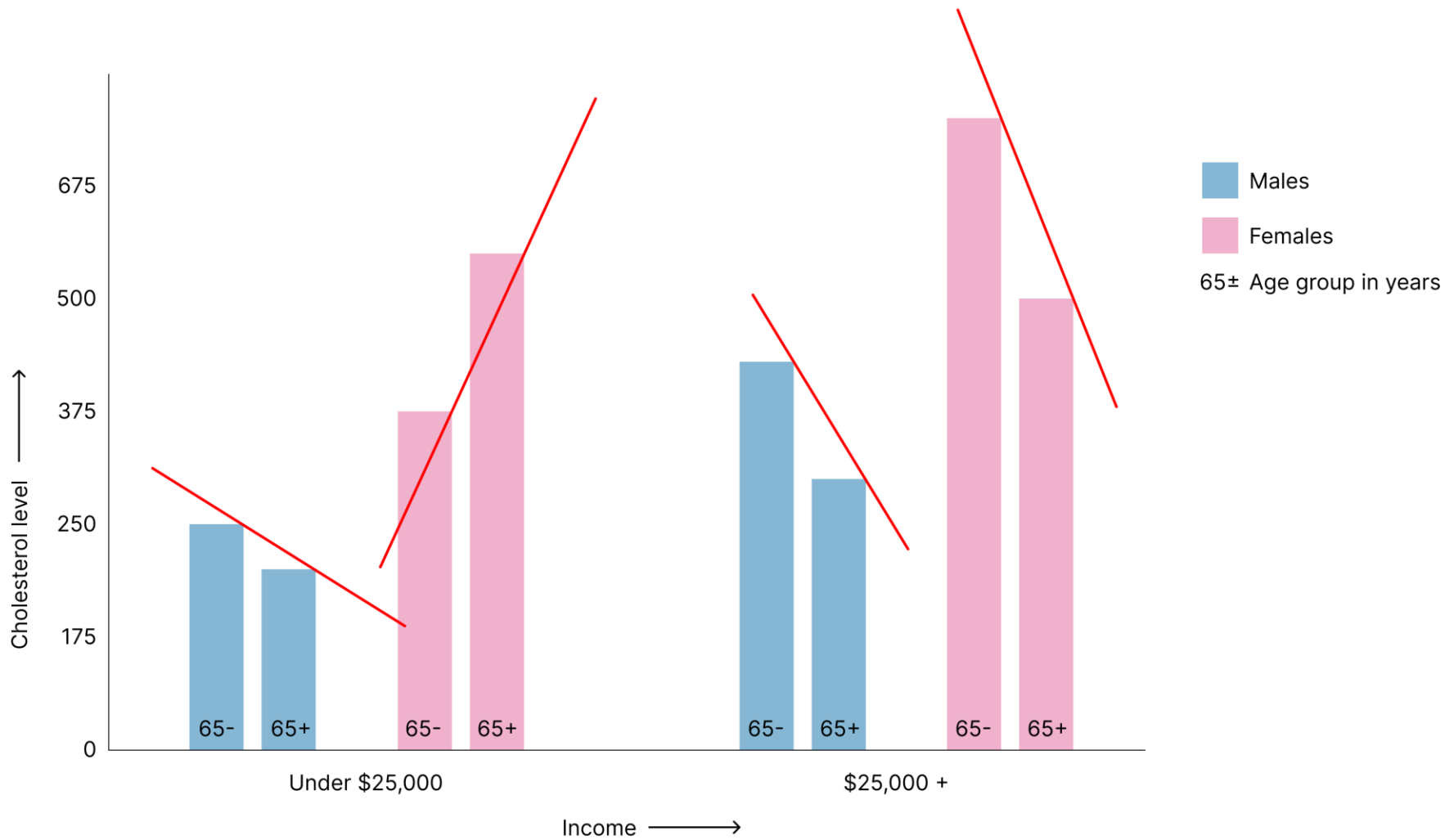
Venkatesh Rajamanickam (@venkatrajam)
venkatra@iitb.ac.in
<http://info-design-lab.github.io>

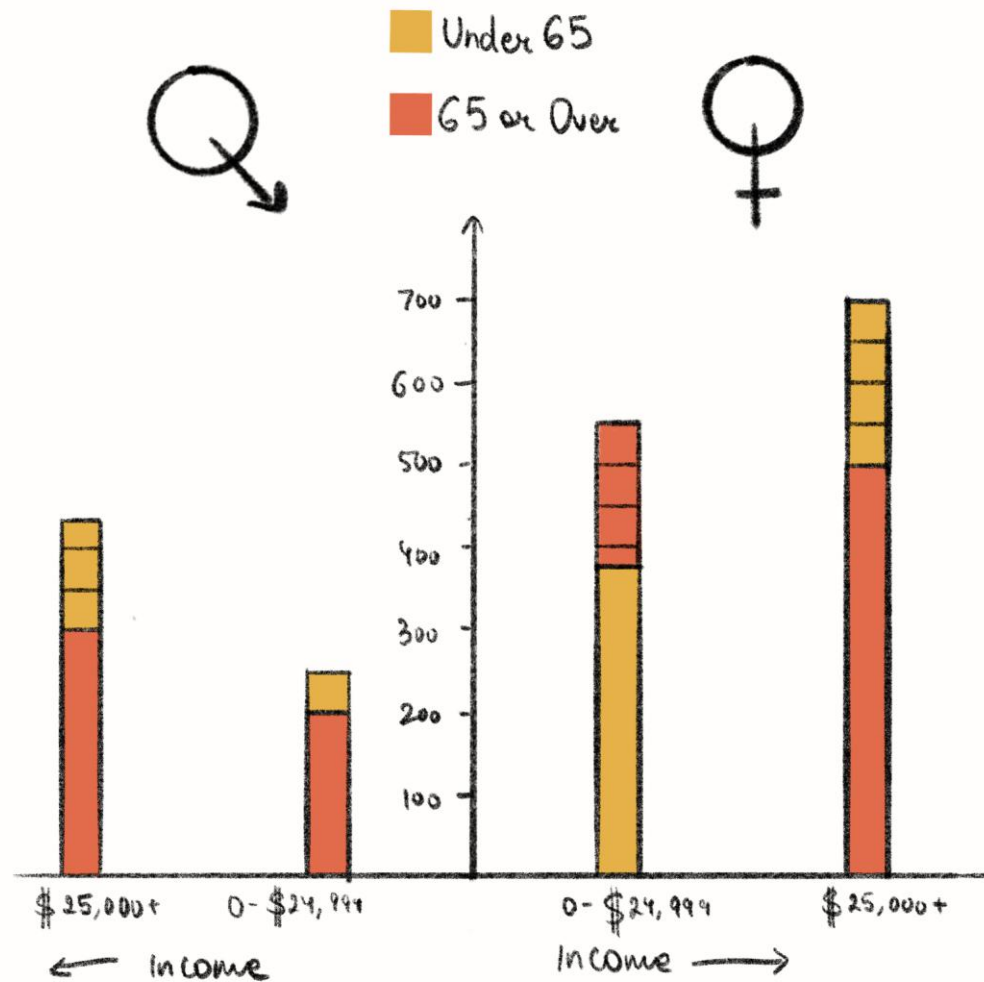
Visualize this!

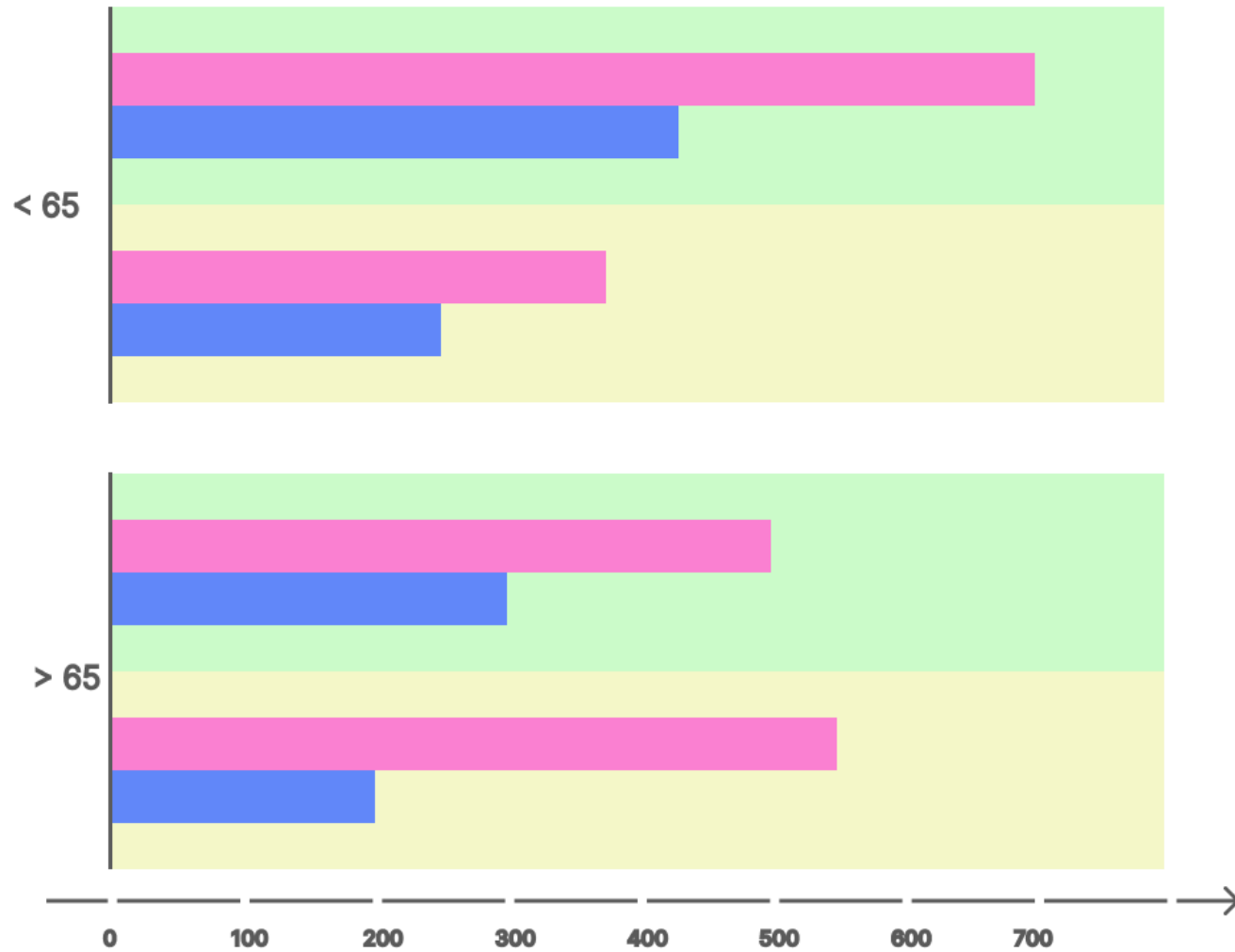
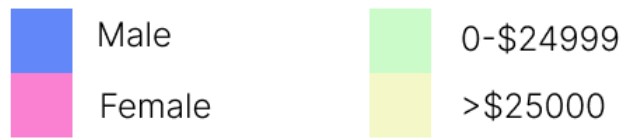
Which gender or income level group shows different effects of age on cholesterol levels?

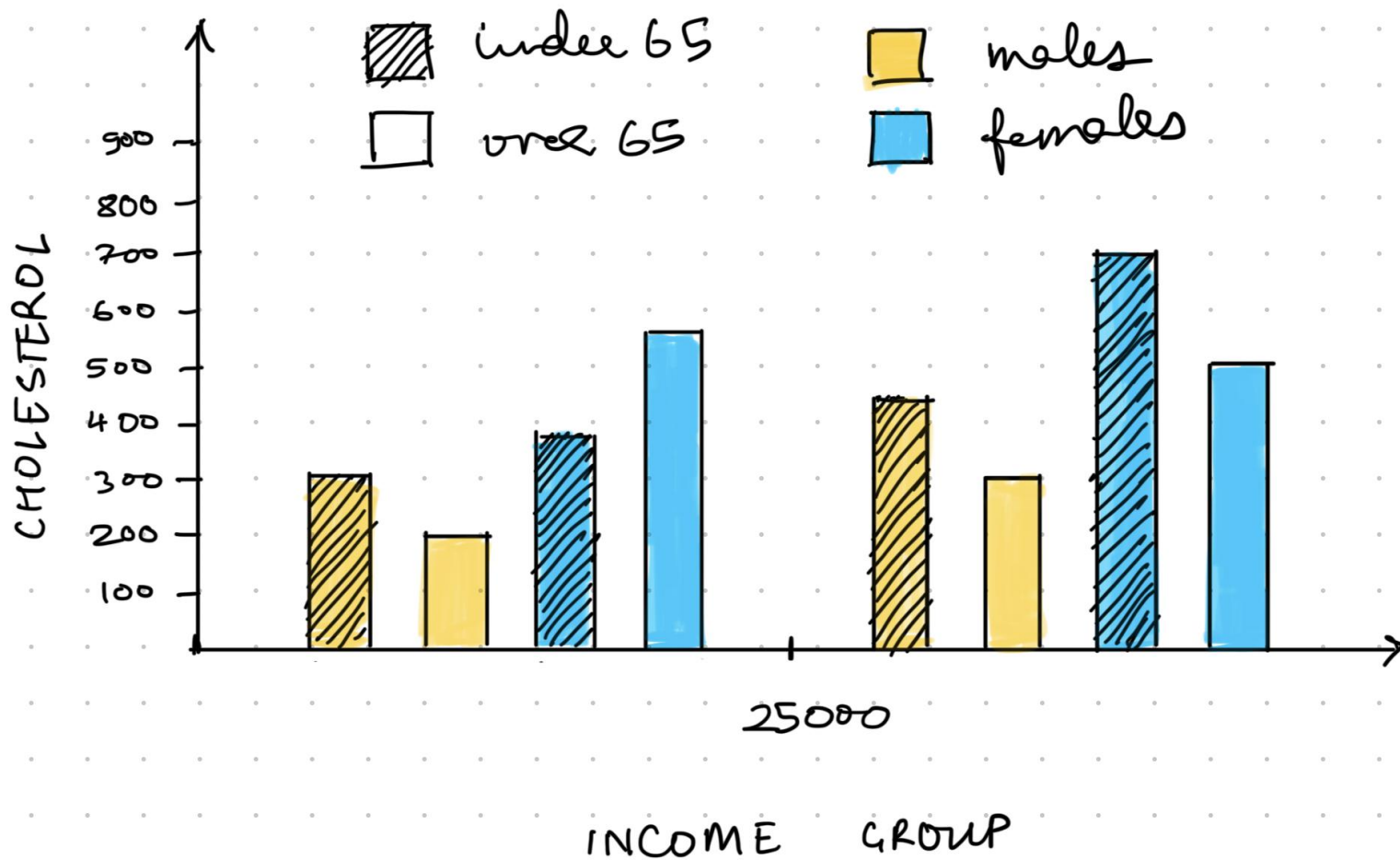
	Males		Females	
Income Group	Under 65	65 or Over	Under 65	65 or Over
0-\$24,999	250	200	375	550
\$25,000+	430	300	700	500

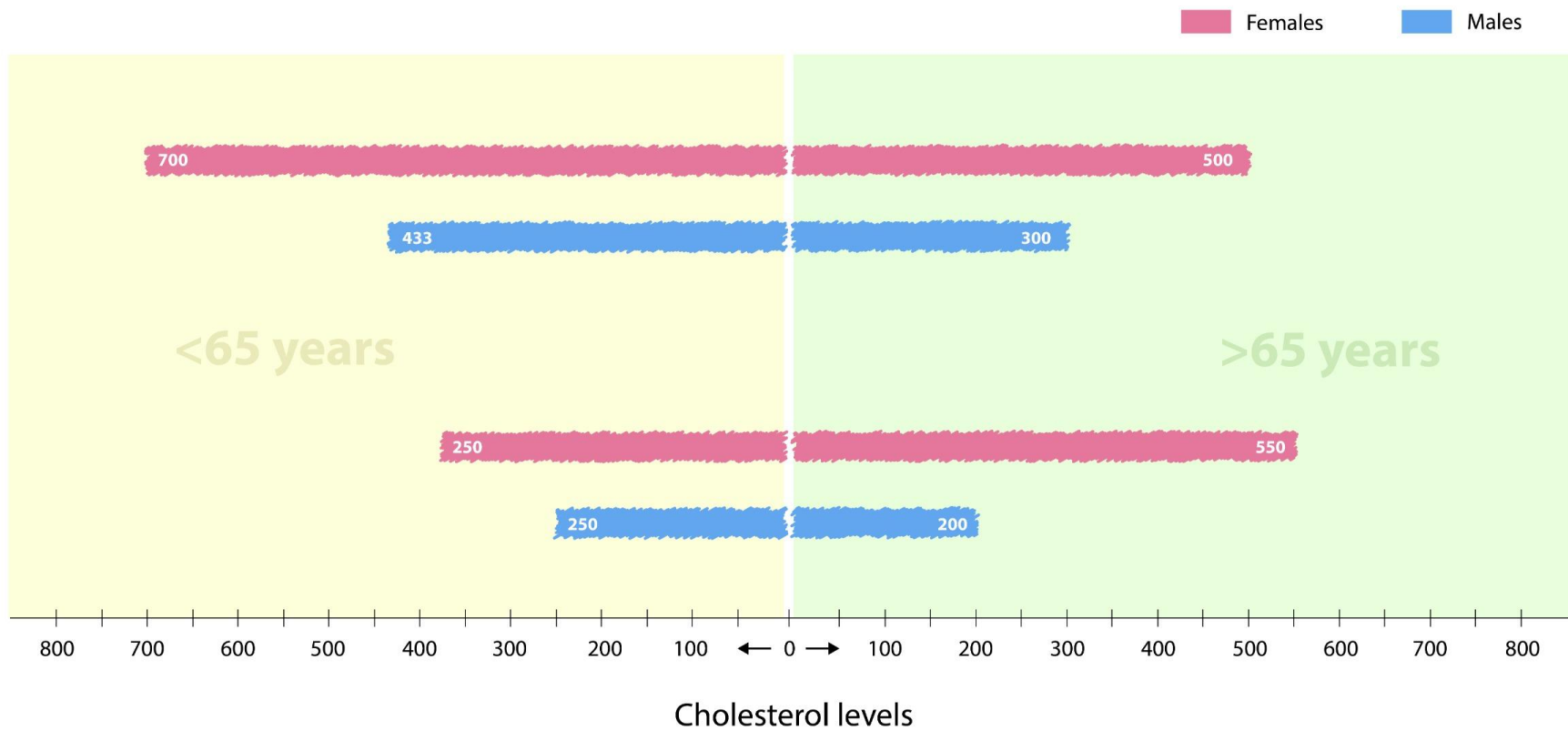




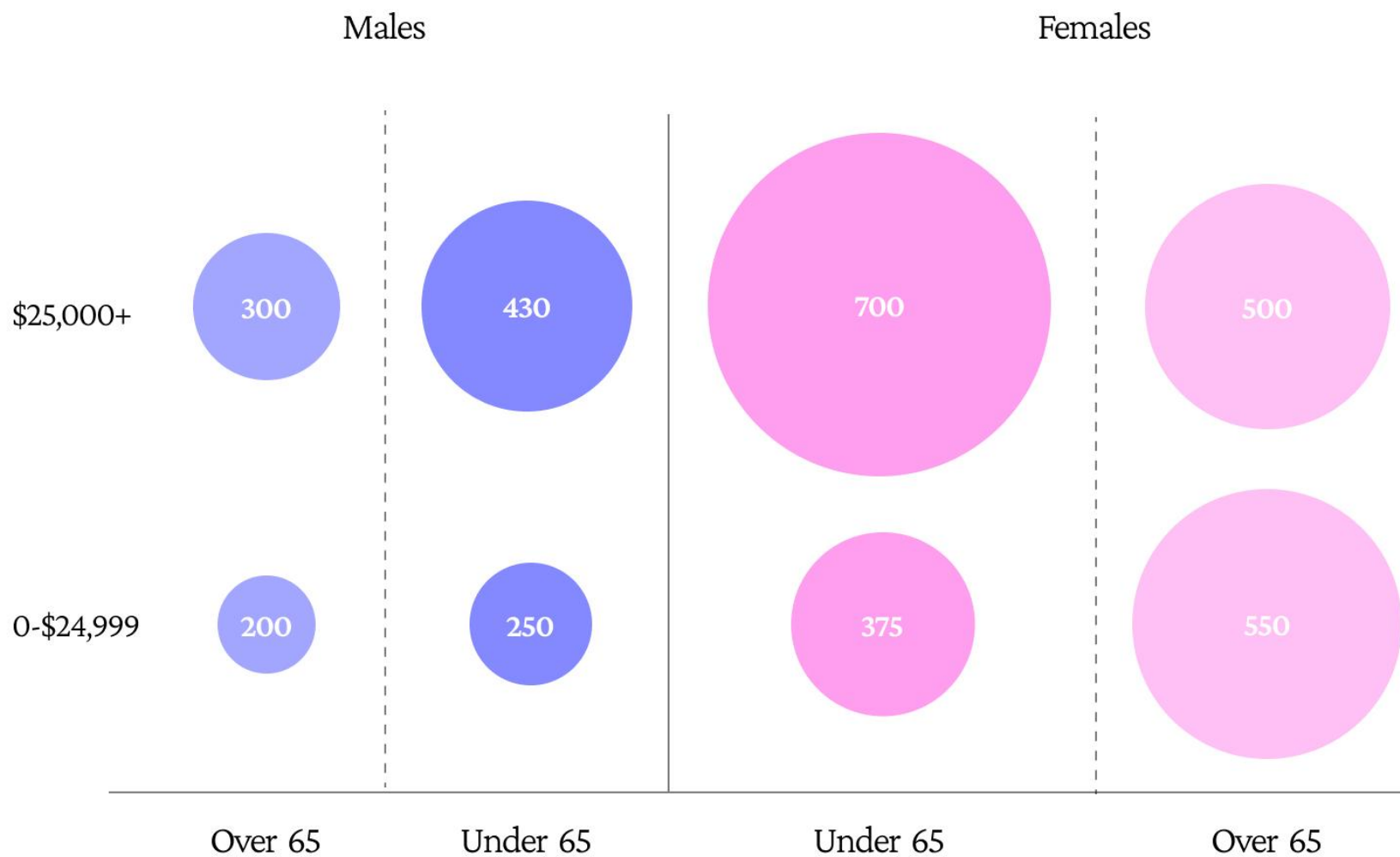


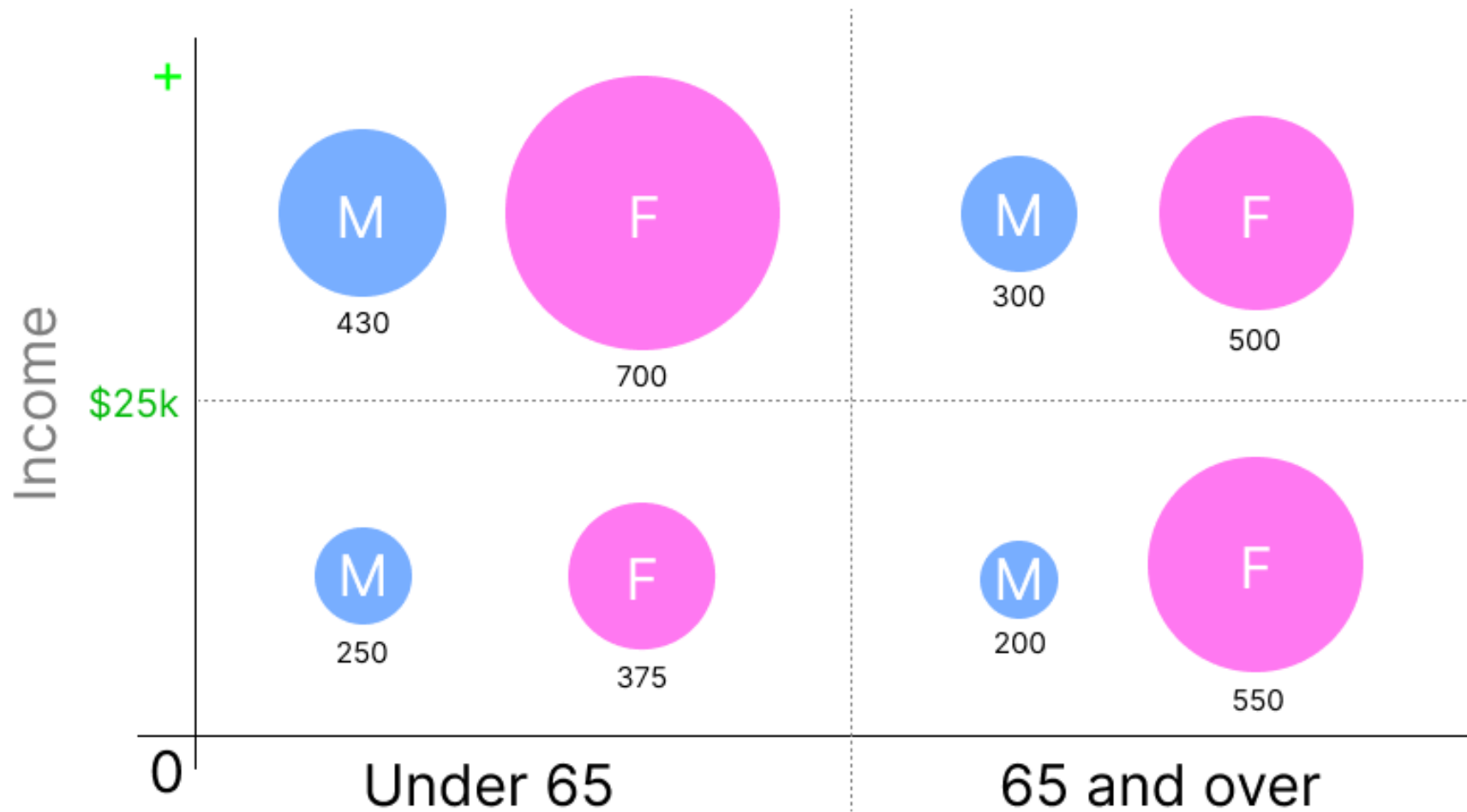






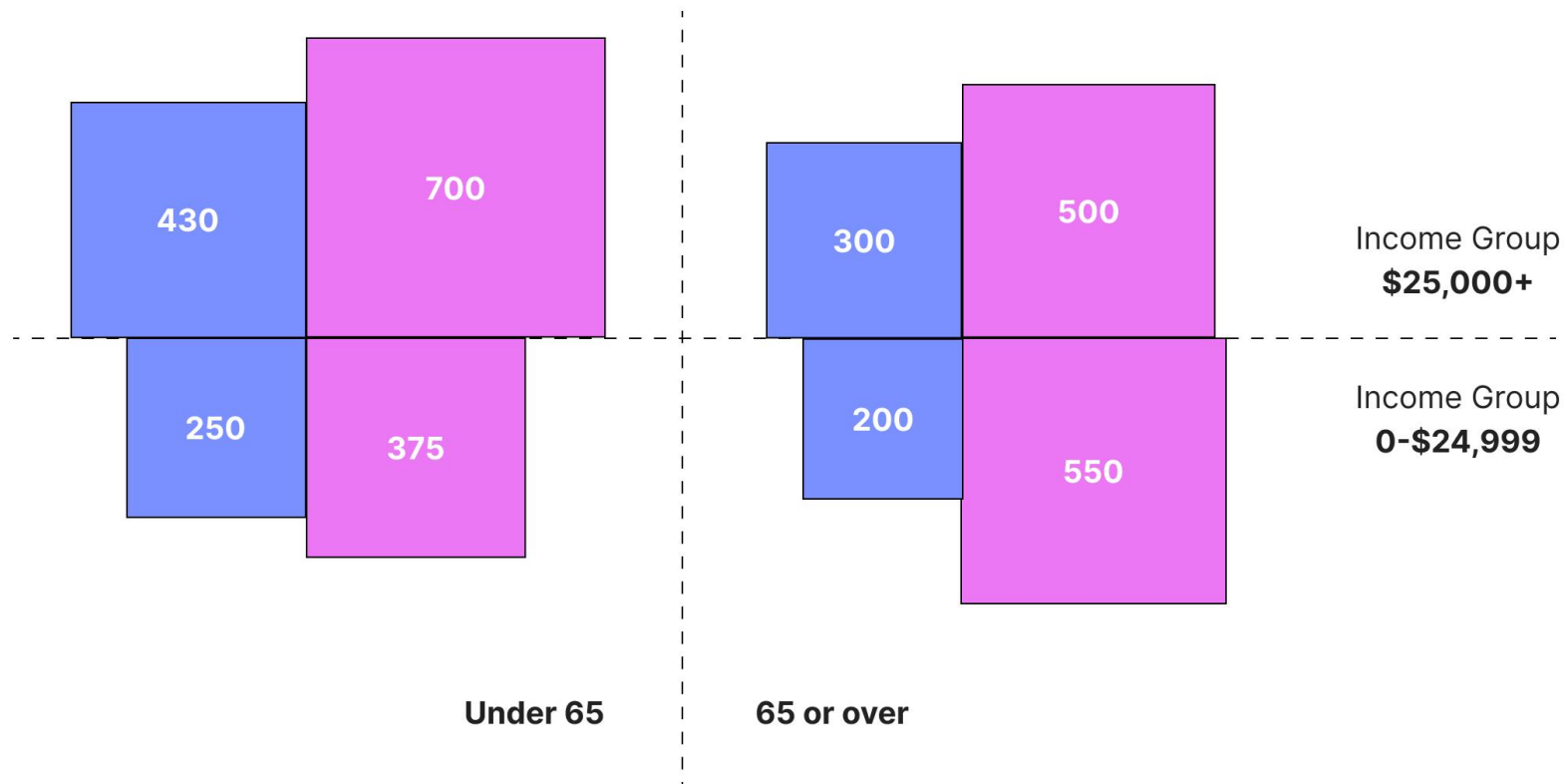
Cholesterol Level among males and females under and over 65 , high and low income



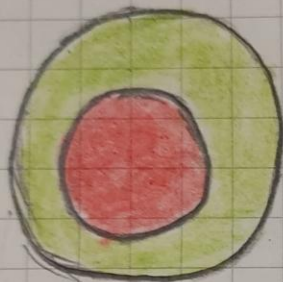


Number of people with cholestrol

Males
Females



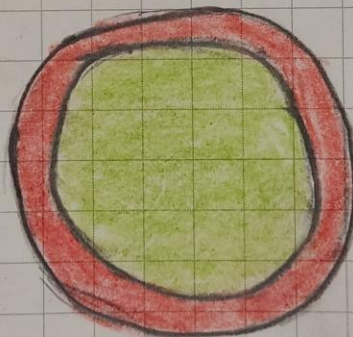
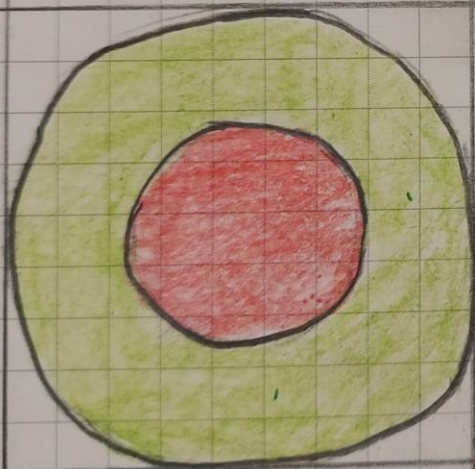
♂



■ \$25,000 +

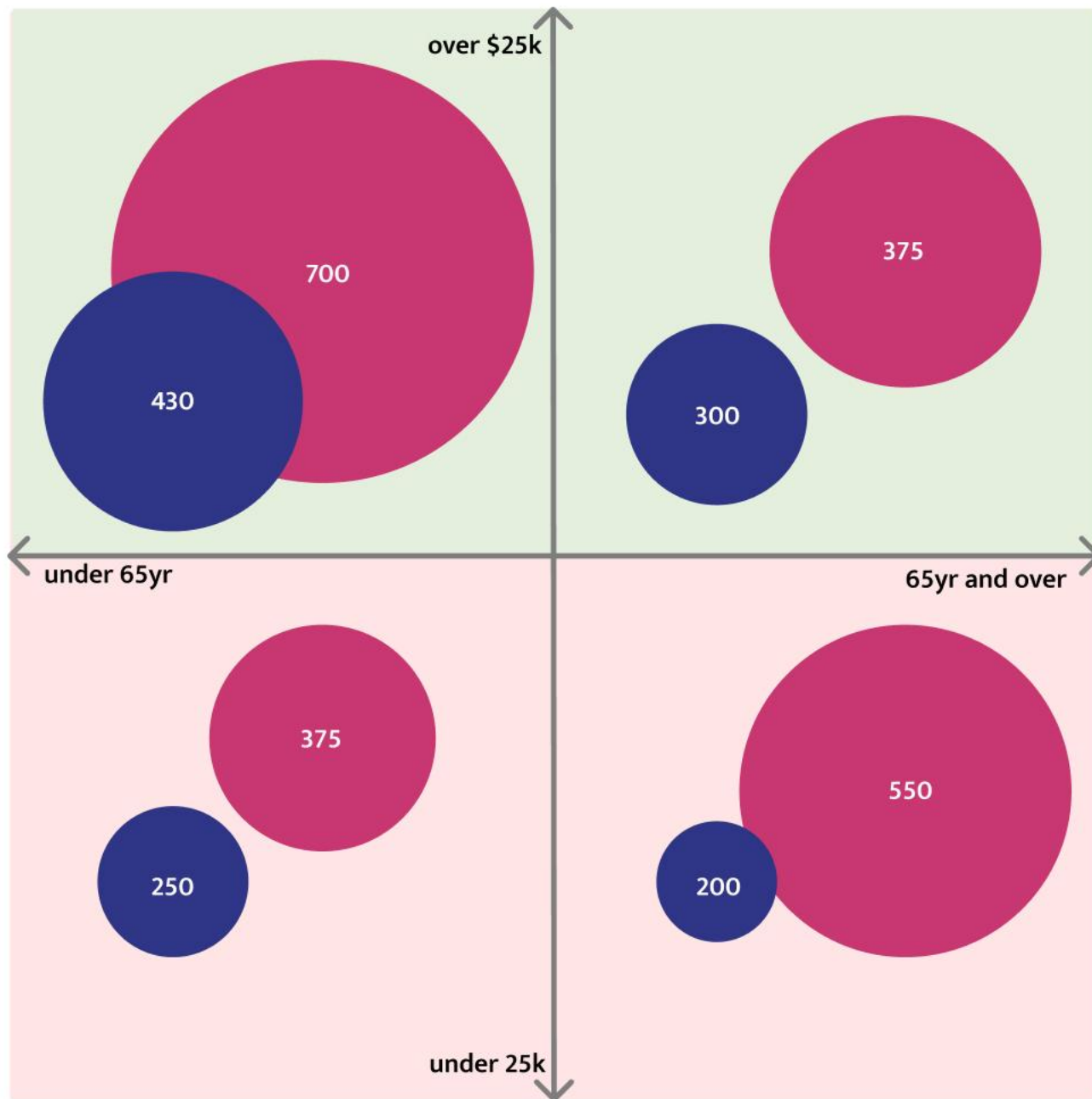
■ \$0 - 24,999

♀



under 65

65 +

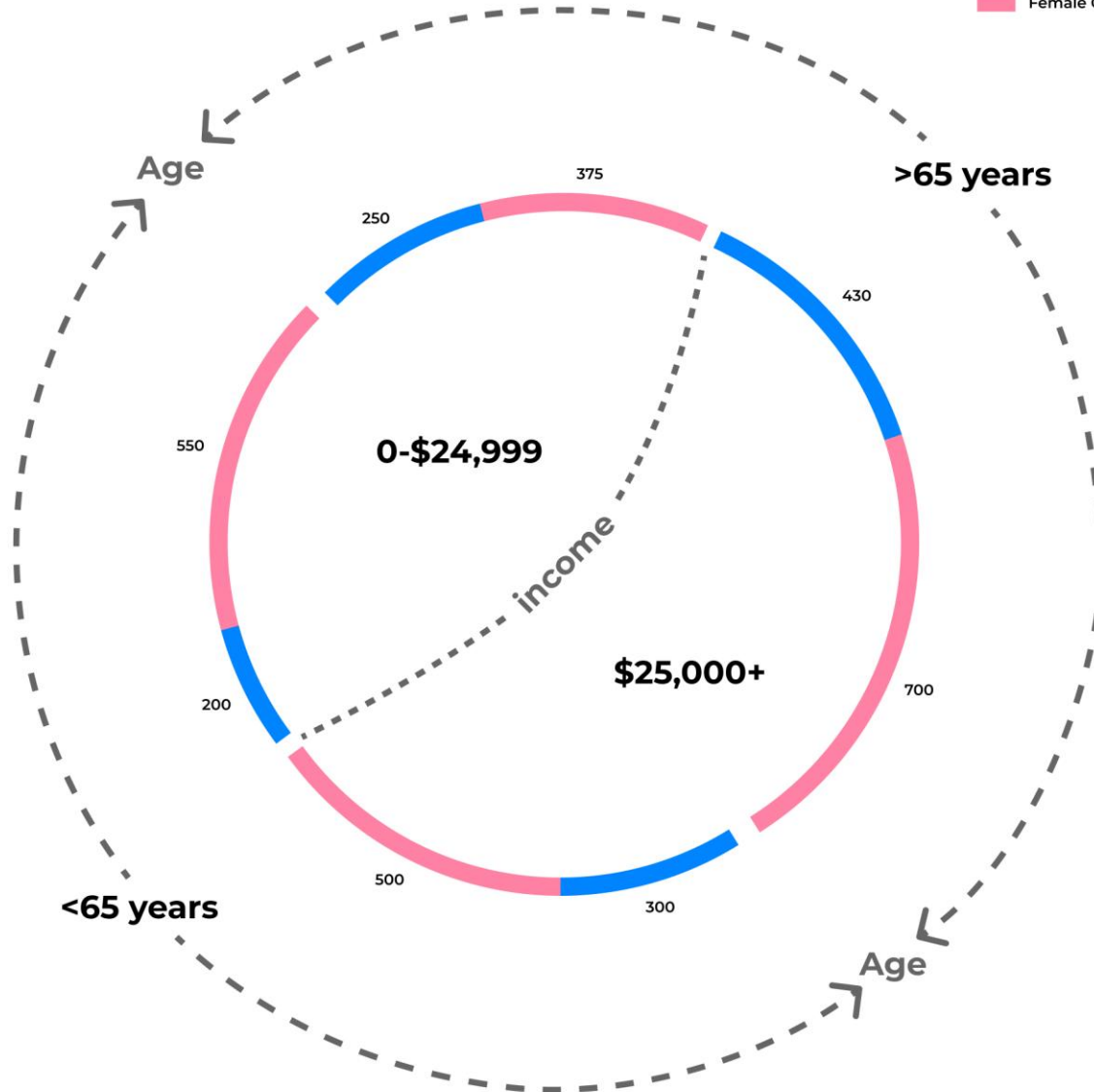


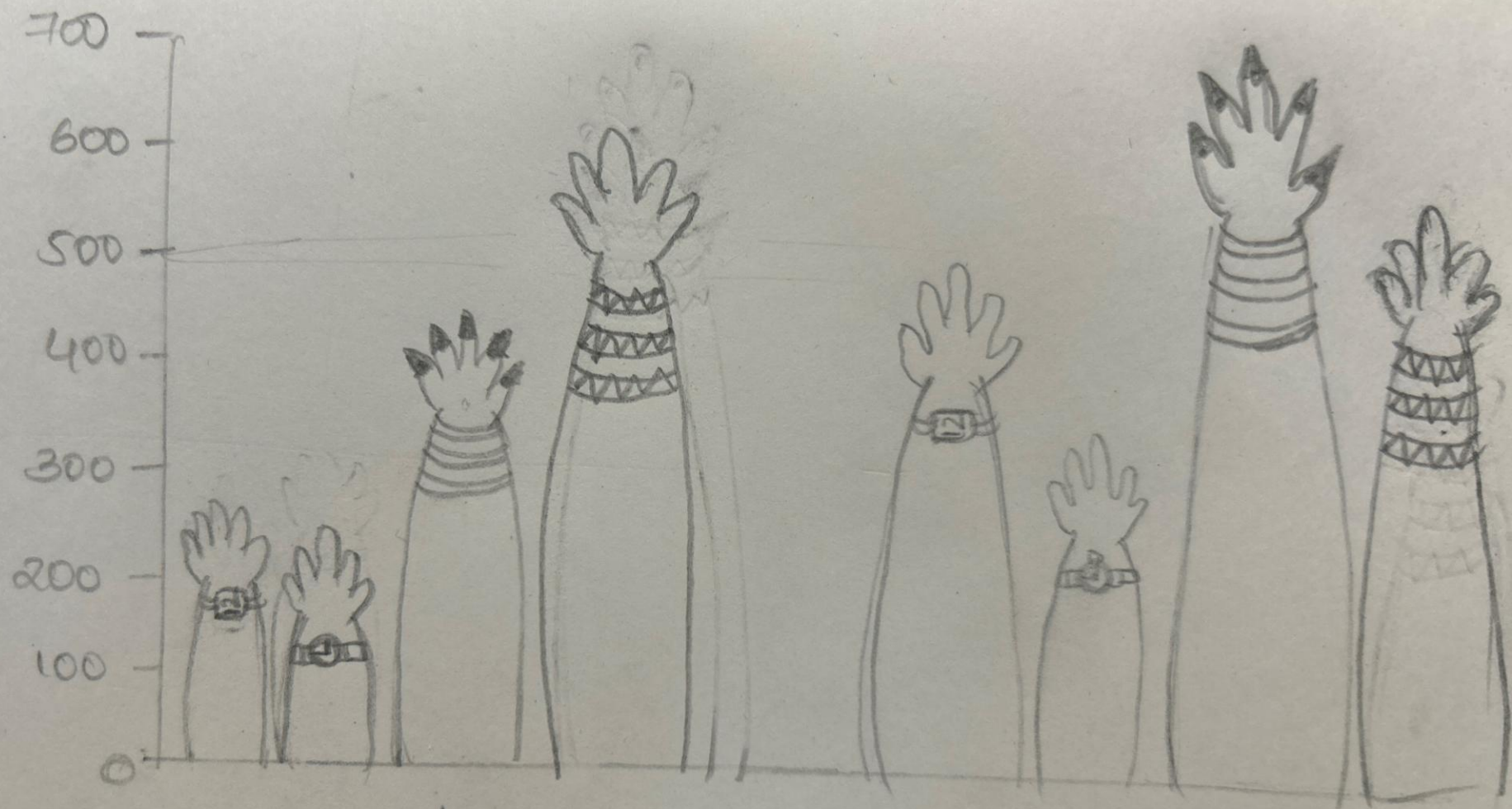
male

female

Cholesterol Level

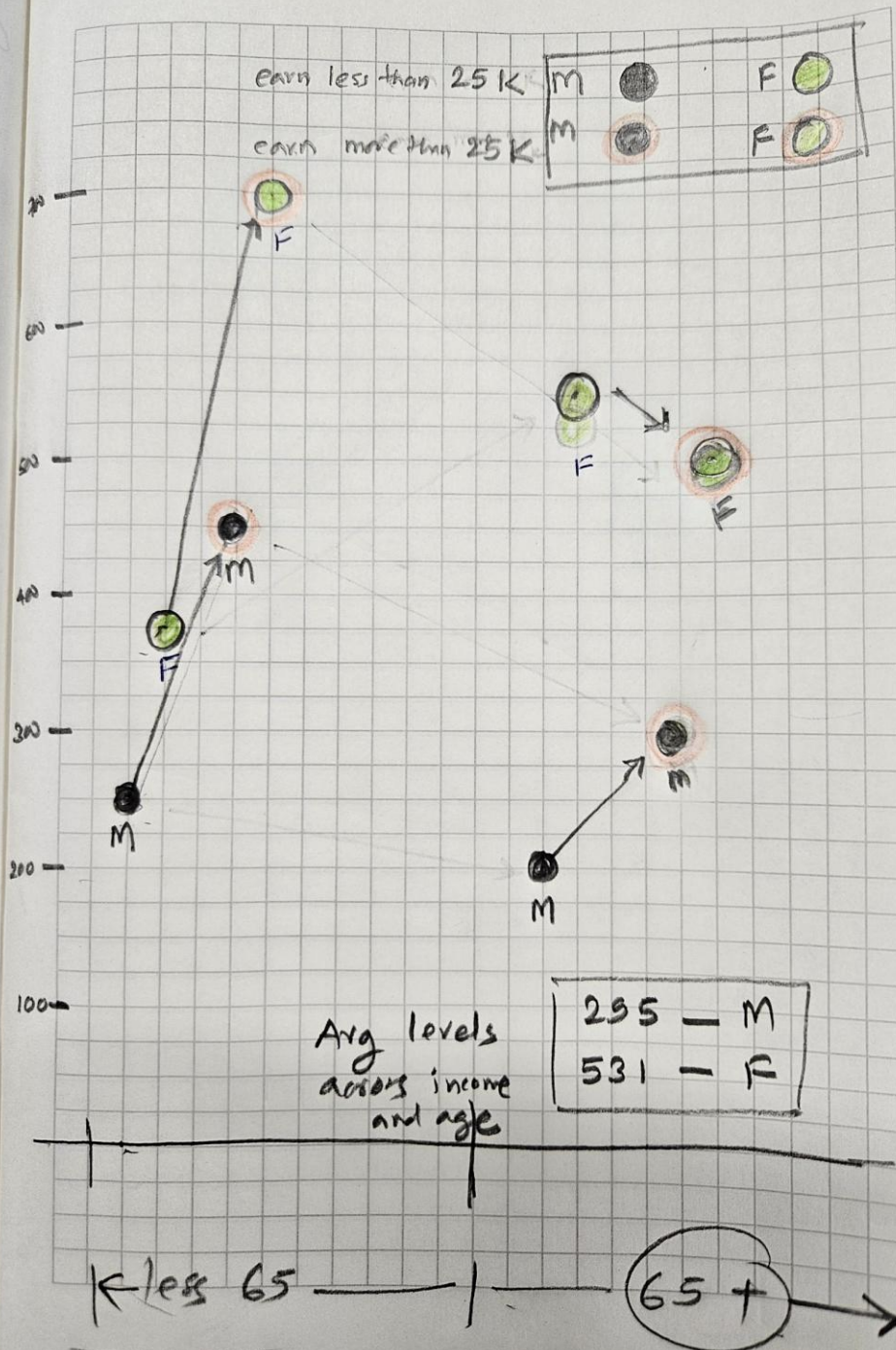
Male Cholesterol Level
Female Cholesterol Level

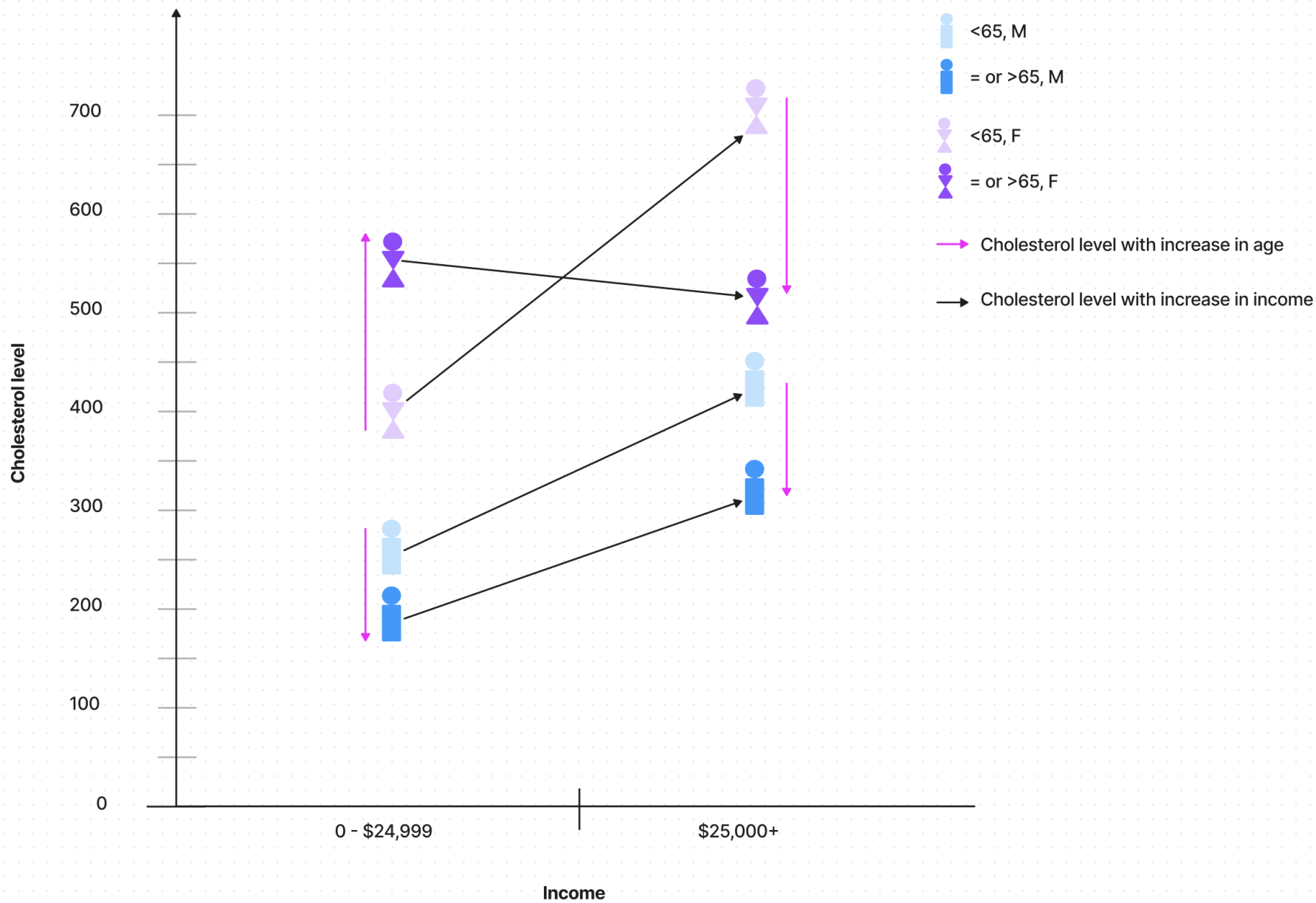




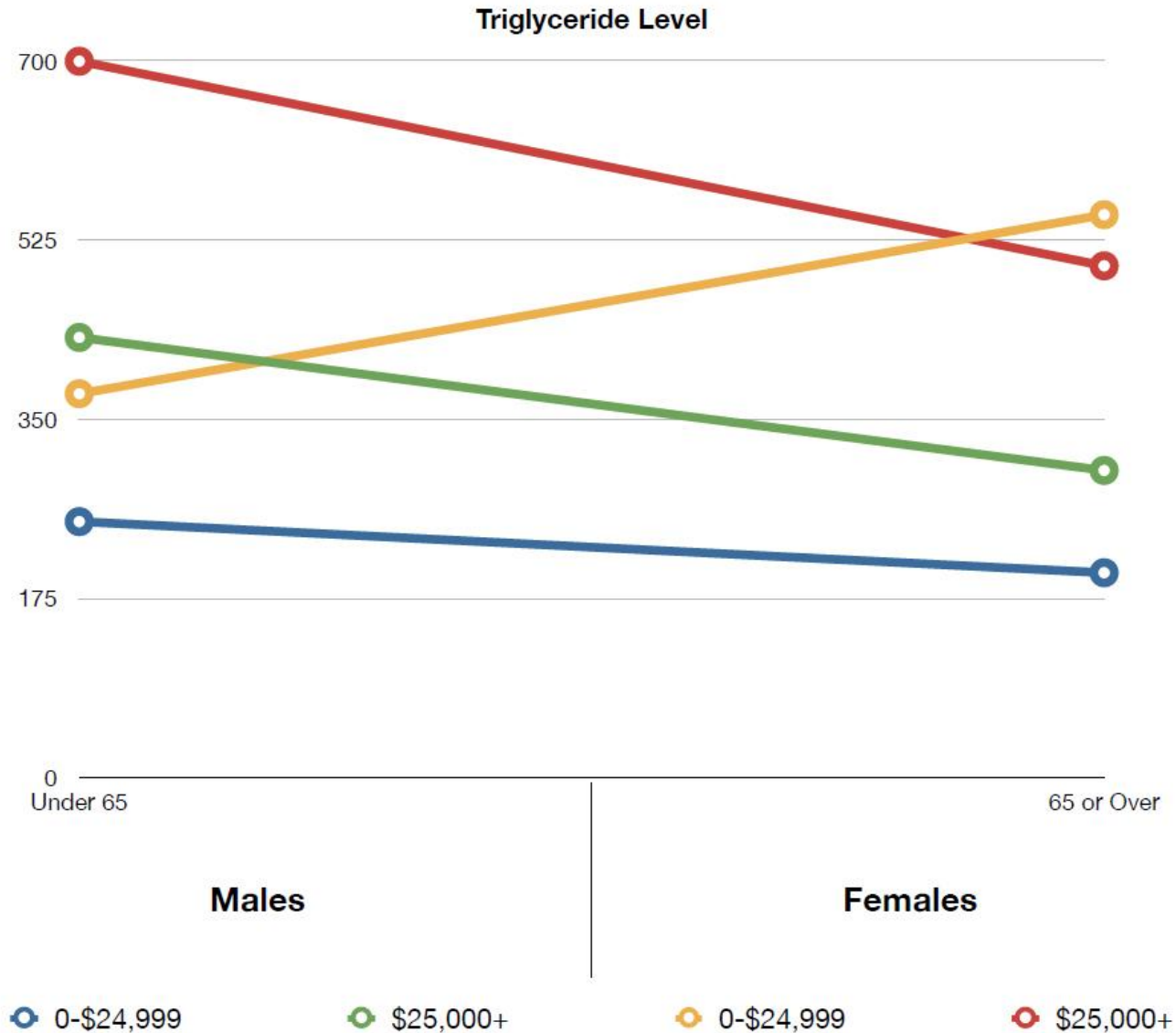
0 - \$24,999

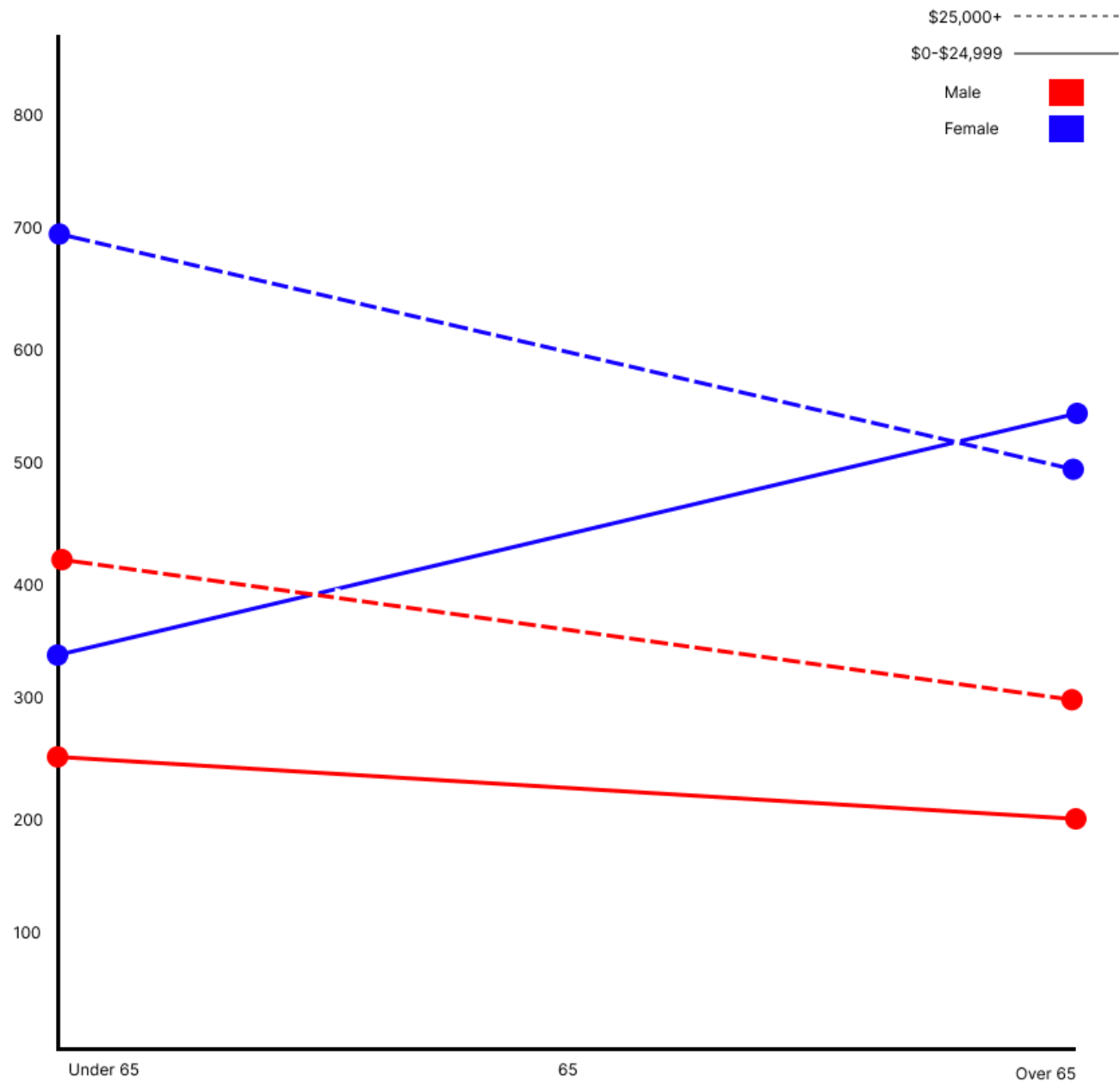
\$25,000 +





Visualize this! - class participation exercise 1





Cholera epidemic 1854 London – Dr Snow

Cholera epidemic 1854 London – Dr Snow



A COURT FOR KING CHOLERA.

Cholera epidemic 1854 London – Dr Snow

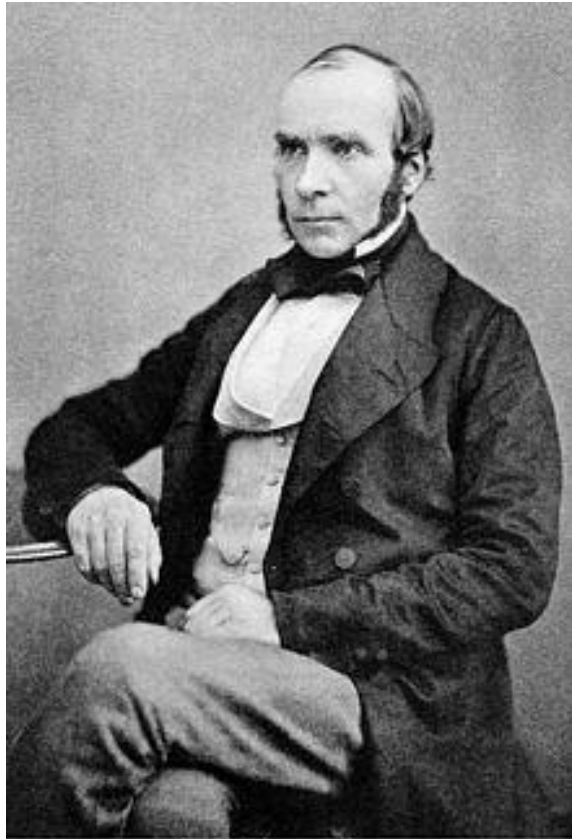


Cholera epidemic 1854 London – Dr Snow



CHOLERA "TRAMPLES THE VICTOR & THE VANQUISHED BOTH."

Cholera epidemic 1854 London – Dr Snow



John Snow

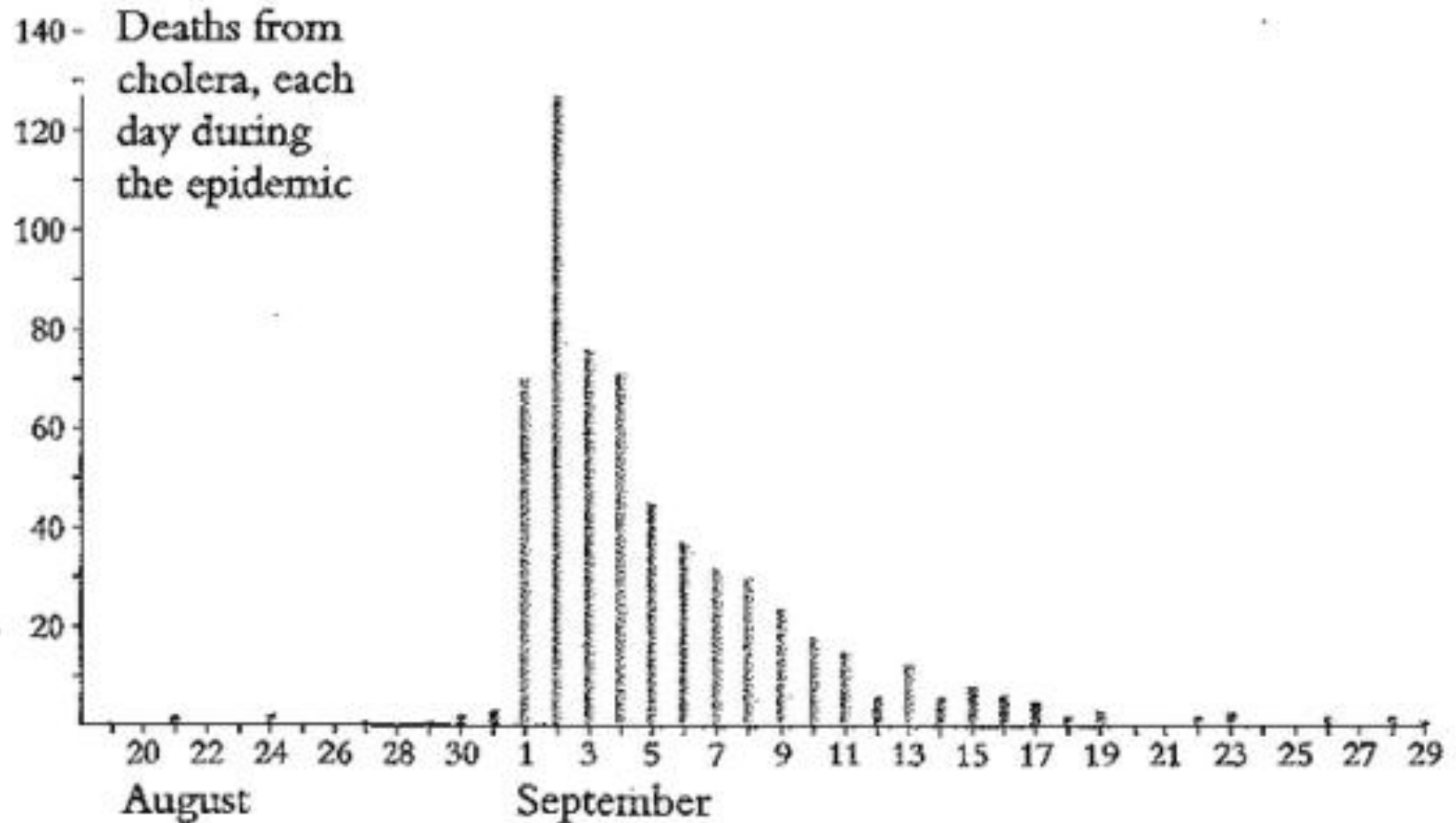




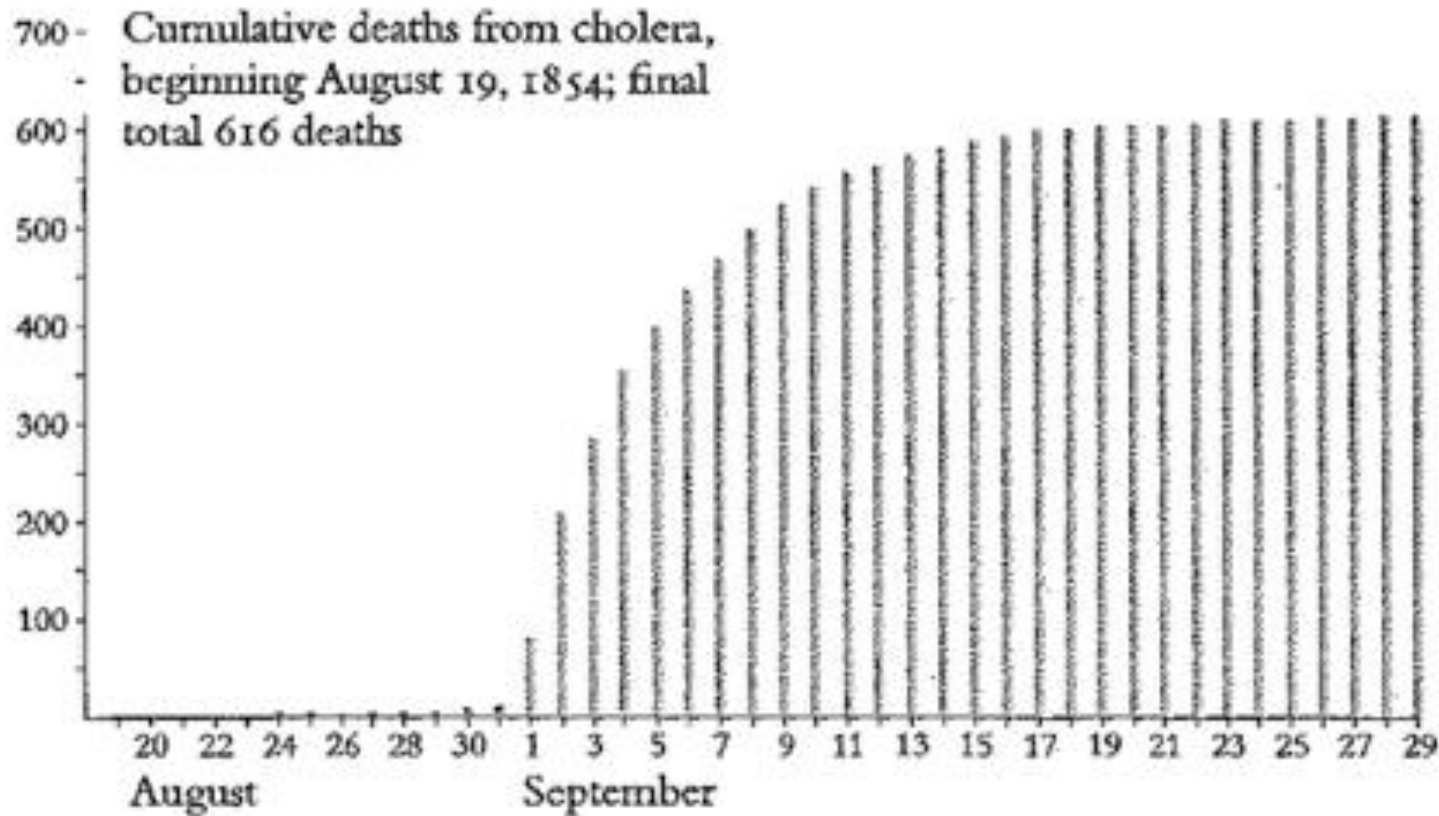
DEATH'S DISPENSARY.

OPEN TO THE POOR, GRATIS, BY PERMISSION OF THE PARISH.

Cholera epidemic 1854 London – Dr Snow



Cholera epidemic 1854 London – Dr Snow



In this case, the task is to create a casual theory about how the disease spread, and time series is a poor match for the task.

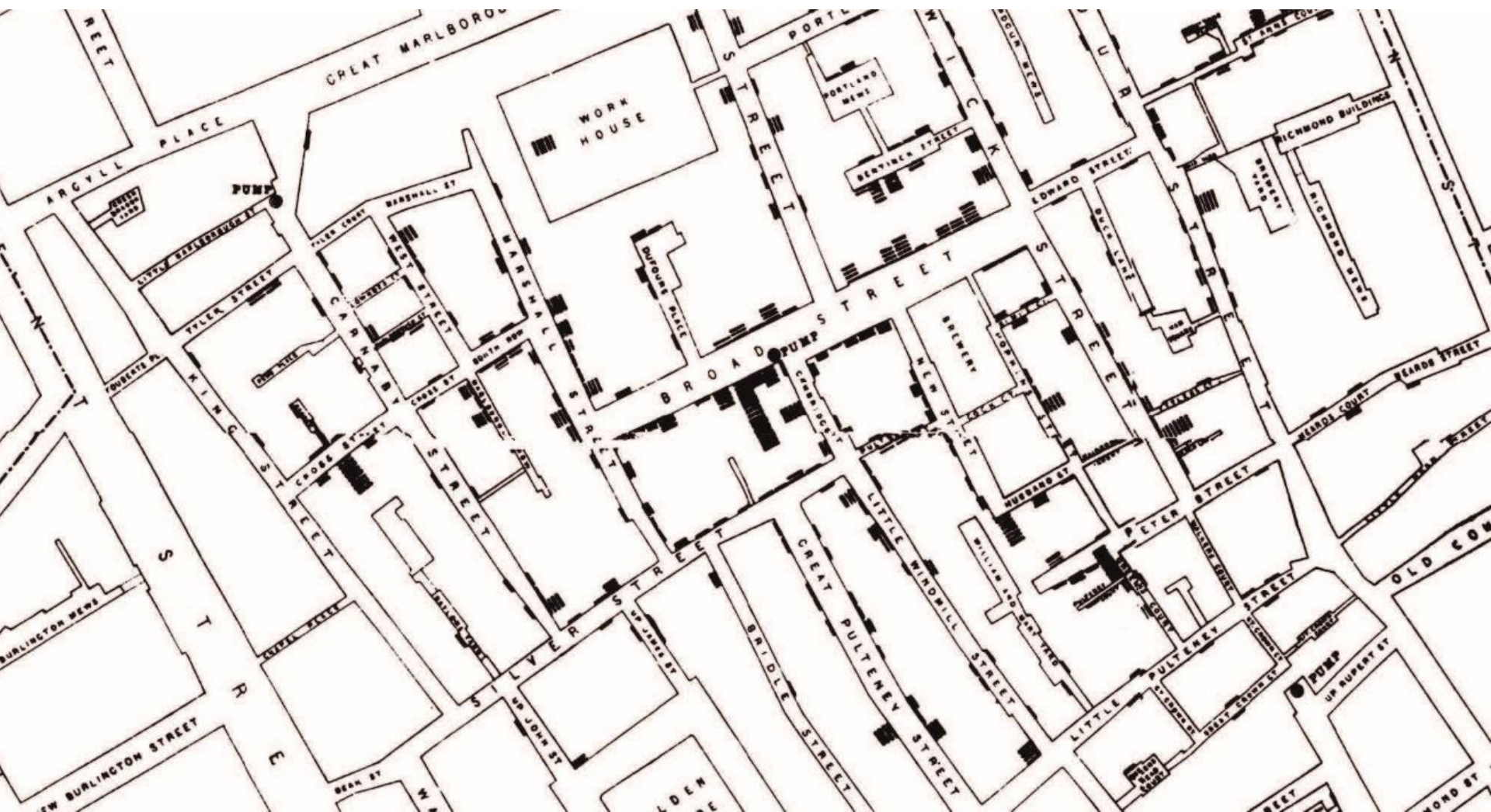
Cholera epidemic 1854 London – Dr Snow

The graphical display was aimed at conveying information about a possible cause-effect relationship.

Snow marked

- deaths from cholera (IIIIII)
- locations of 11 community water pumps.

Cholera epidemic 1854 London – Dr Snow



CHOLERA AND **WATER.**

BOARD OF WORKS

FOR THE LIMEHOUSE DISTRICT,
Comprising Limehouse, Ratcliff, Shadwell,
and Wapping.

The **INHABITANTS** of the District within
which **CHOLERA IS PREVAILING**, are
earnestly advised

NOT TO DRINK ANY WATER
WHICH HAS NOT
PREVIOUSLY BEEN BOILED.

Fresh Water ought to be Boiled every
Morning for the day's use, and what
remains of it ought to be thrown away
at night. The Water ought not to stand
where any kind of dirt can get into it,
and great care ought to be given to see
that Water Butts and Cisterns are free
from dirt.

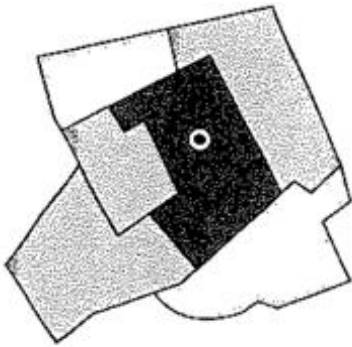
BY ORDER,

THOS. W. RATCLIFF,
CLERK OF THE BOARD.

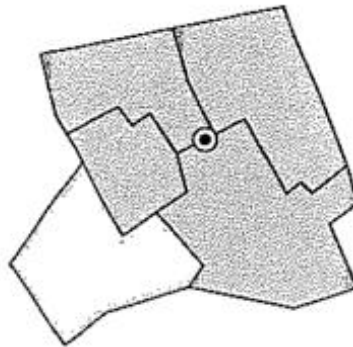
Cholera epidemic 1854 London – Dr Snow

Snow drew a “dot map”, which marks each individual death

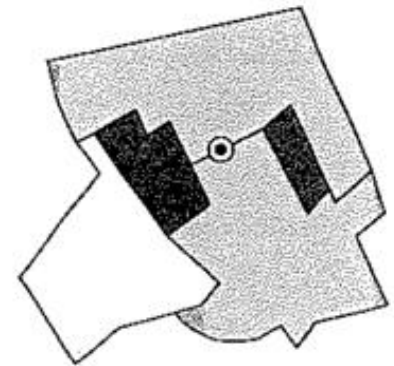
Why not aggregate?



In this aggregation of individual deaths into six areas, the greatest number is concentrated at the Broad Street pump.



Using different geographic subdivisions, the cholera numbers are nearly the same in four of the five areas.

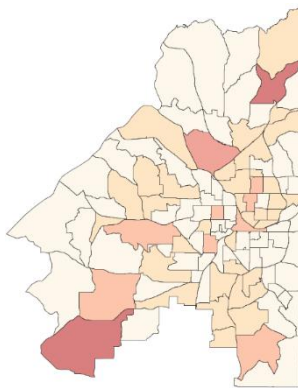


In this aggregation of the deaths, the two areas with the most deaths do not even include the infected pump!

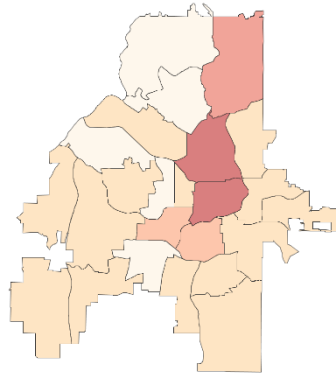
Cholera epidemic 1854 London – Dr Snow

Modifiable Area Unit Problem

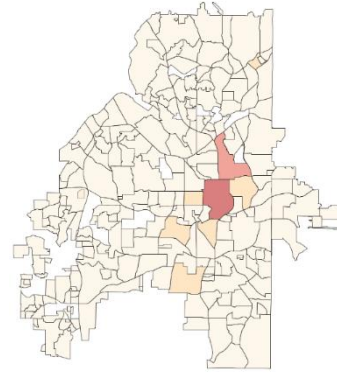
Alternative district shapes create different distributions for point-based data



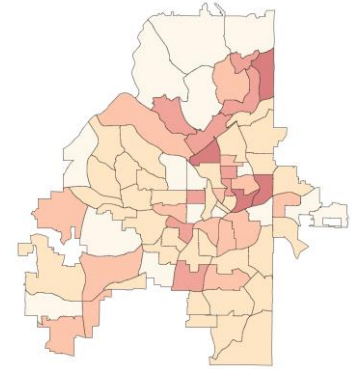
Census



Wards



Neighborhoods



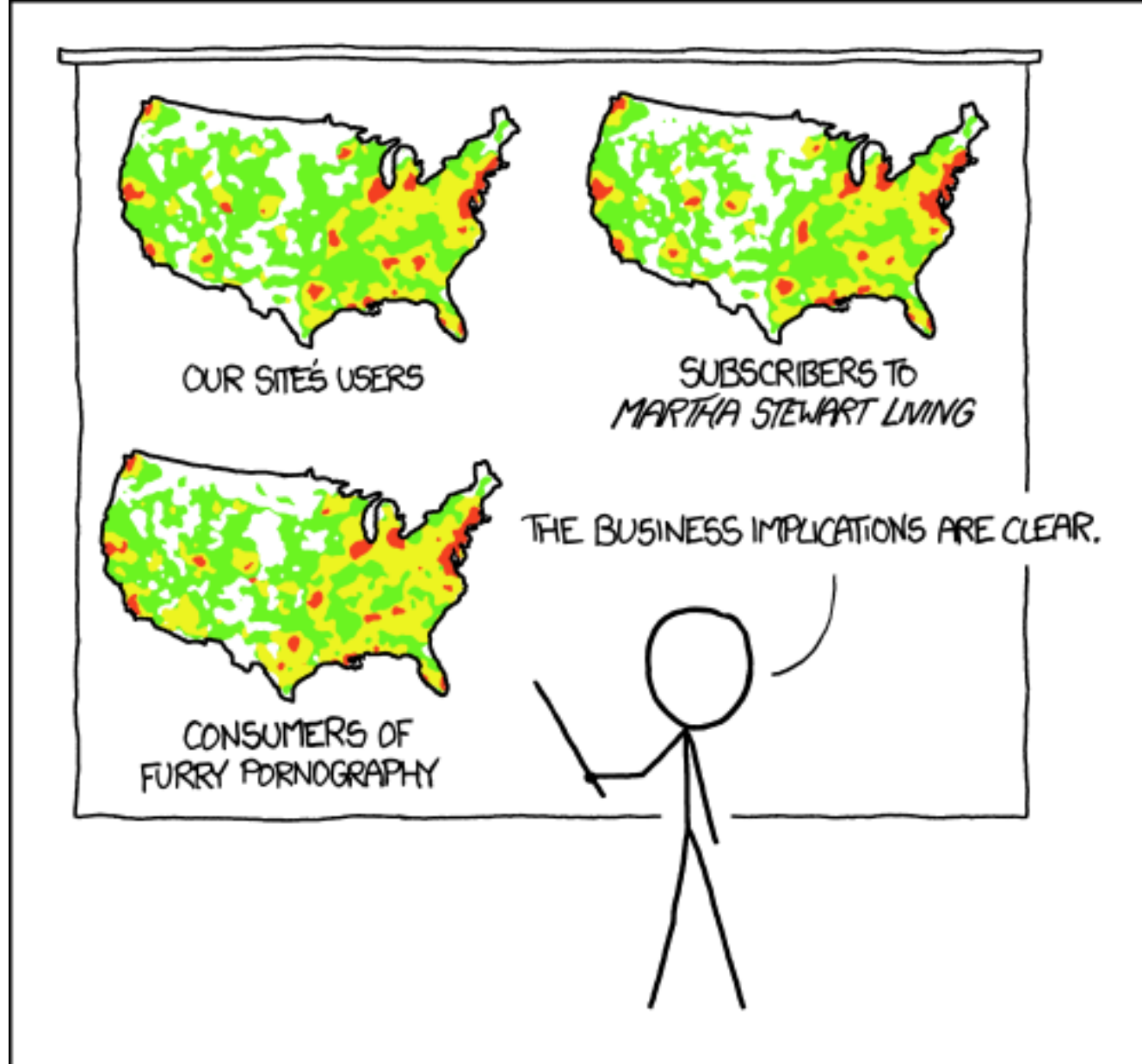
Police Beats

Cholera epidemic 1854 London – Dr Snow

But what is the flip side of Snow's Display?

The dot map does not take into account the number of people living in an area

(an area may be free of cases because it is not populated, maybe more people lived near Broad Street pump?)

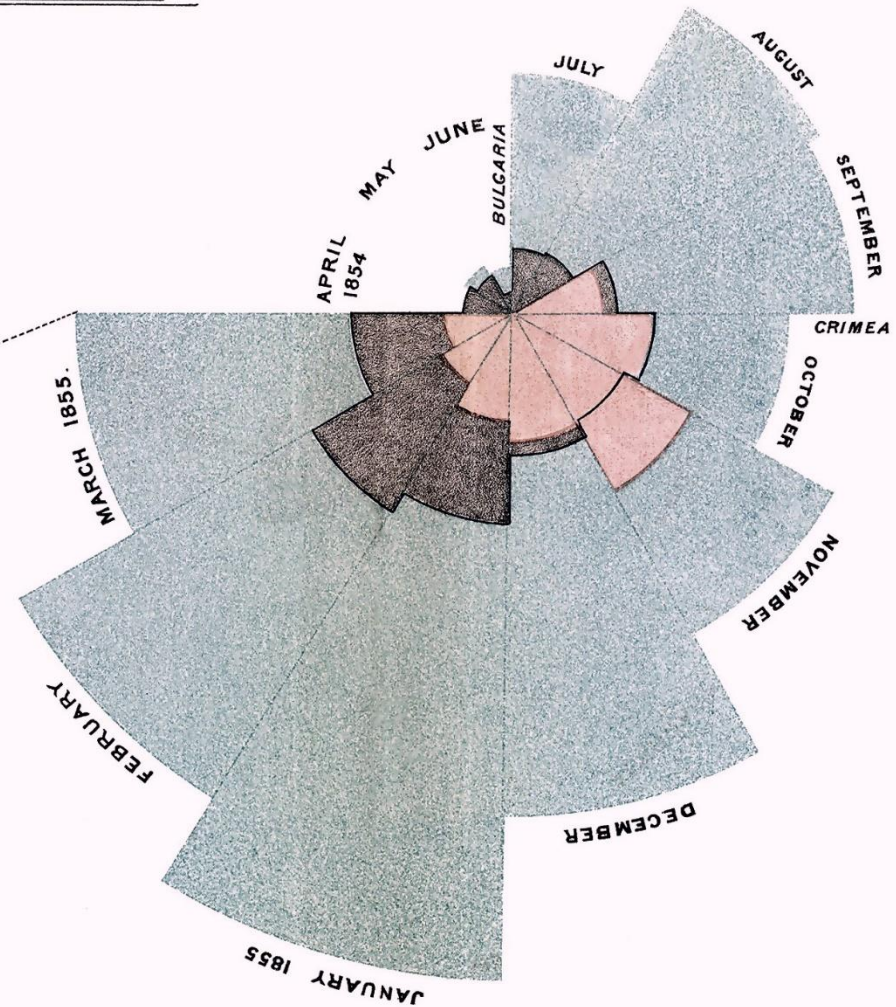


PET PEEVE #208:
GEOGRAPHIC PROFILE MAPS WHICH ARE
BASICALLY JUST POPULATION MAPS

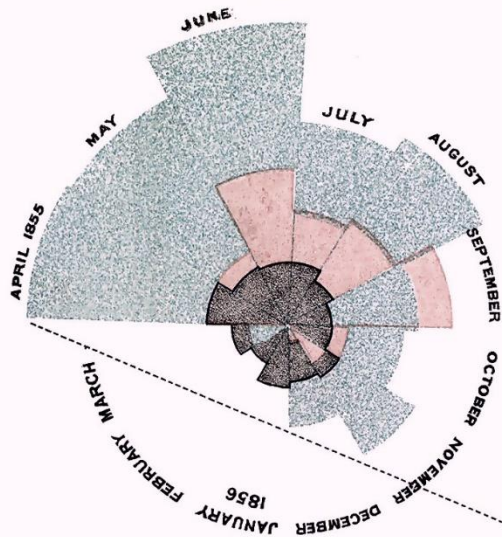
Florence Nightingale on Crimean War. 1858

DIAGRAM OF THE CAUSES OF MORTALITY IN THE ARMY IN THE EAST.

1.
APRIL 1854 to MARCH 1855.



2.
APRIL 1855 to MARCH 1856.



The Areas of the blue, red, & black wedges are each measured from the centre as the common vertex.

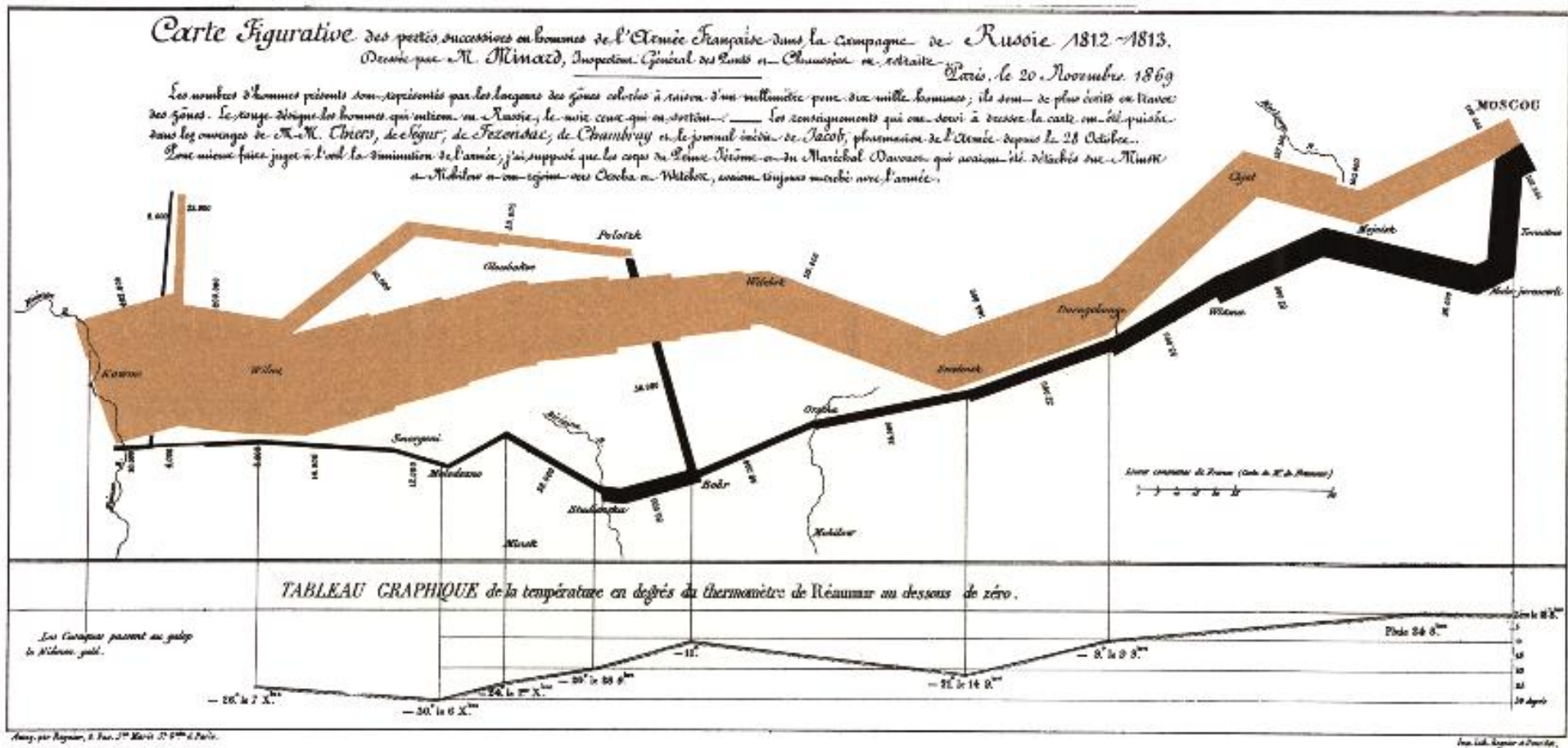
The blue wedges measured from the centre of the circle represent area for area the deaths from Preventable or Mitigable Zymotic diseases; the red wedges measured from the centre the deaths from wounds; & the black wedges measured from the centre the deaths from all other causes.

The black line across the red triangle in Nov. 1854 marks the boundary of the deaths from all other causes during the month.

In October 1854, & April 1855, the black area coincides with the red; in January & February 1856, the blue coincides with the black.

The entire areas may be compared by following the blue, the red & the black lines enclosing them.

Napoleon's 1812 Russian campaign – Charles Minard



REFERENCE

- District Railway.
- London Electric Railway.
- City and South London Rly.
- Central London Railway.
- Metropolitan Railway.
- Metropolitan Railway.
- Interchange Stations.
- Piccadilly Railway Extensions under construction.

UNDERGROUND RAILWAYS OF LONDON

London Underground Map 1931 – Henry Beck



Periodic Table 1869 – Dimitri Mendeleev

Ueber die Beziehungen der Eigenschaften zu den Atomgewichten der Elemente. Von D. Mendelejeff. — Ordnet man Elemente nach zunehmenden Atomgewichten in verticale Reihen so, dass die Horizontalreihen analoge Elemente enthalten, wieder nach zunehmendem Atomgewicht geordnet, so erhält man folgende Zusammenstellung, aus der sich einige allgemeinere Folgerungen ableiten lassen.

			Ti = 50	Zr = 90	? = 180
			V = 51	Nb = 94	Ta = 182
			Cr = 52	Mo = 96	W = 186
			Mn = 55	Rh = 104,4	Pt = 197,4
			Fe = 56	Ru = 104,4	Ir = 198
		Ni =	Co = 59	Pd = 106,6	Os = 199
			Cu = 63,4	Ag = 108	Hg = 200
H = 1	Be = 9,4	Mg = 24	Zn = 65,2	Cd = 112	
	B = 11	Al = 27,4	? = 68	Ur = 116	Au = 197?
	C = 12	Si = 28	? = 70	Sn = 118	
	N = 14	P = 31	As = 75	Sb = 122	Bi = 210?
	O = 16	S = 32	Se = 79,4	Te = 128?	
	F = 19	Cl = 35,5	Br = 80	J = 127	
Li = 7	Na = 23	K = 39	Rb = 85,4	Cs = 133	Tl = 204
		Ca = 40	Sr = 87,6	Ba = 137	Pb = 207
		? = 45	Ce = 92		
		?Er = 56	La = 94		
		?Yt = 60	Di = 95		
		?In = 75,6	Th = 118?		

Periodic Table 1950 – Henry Hubbard

PERIODIC CHART OF THE ATOMS

The Atoms Grouped According to the Number of Outer [Valence] Electrons

Henry D. Hubbard
William F. Meggers

Planetary electrons in the completed shells
Total Atom No. = $2(1^2 \cdot 2^2 \cdot 3^2 \cdot 4^2 \cdot 5^2 \cdot 6^2 \cdot 7^2)$

End of the 1st Period
 $2 = 2$

End of the 2nd Period
 $10 = 2 \cdot 8$

End of the 3rd Period
 $18 = 2 \cdot 8 \cdot 8$

End of the 4th Period
 $36 = 2 \cdot 8 \cdot 18 \cdot 8$

End of the 5th Period
 $54 = 2 \cdot 8 \cdot 18 \cdot 18 \cdot 8$

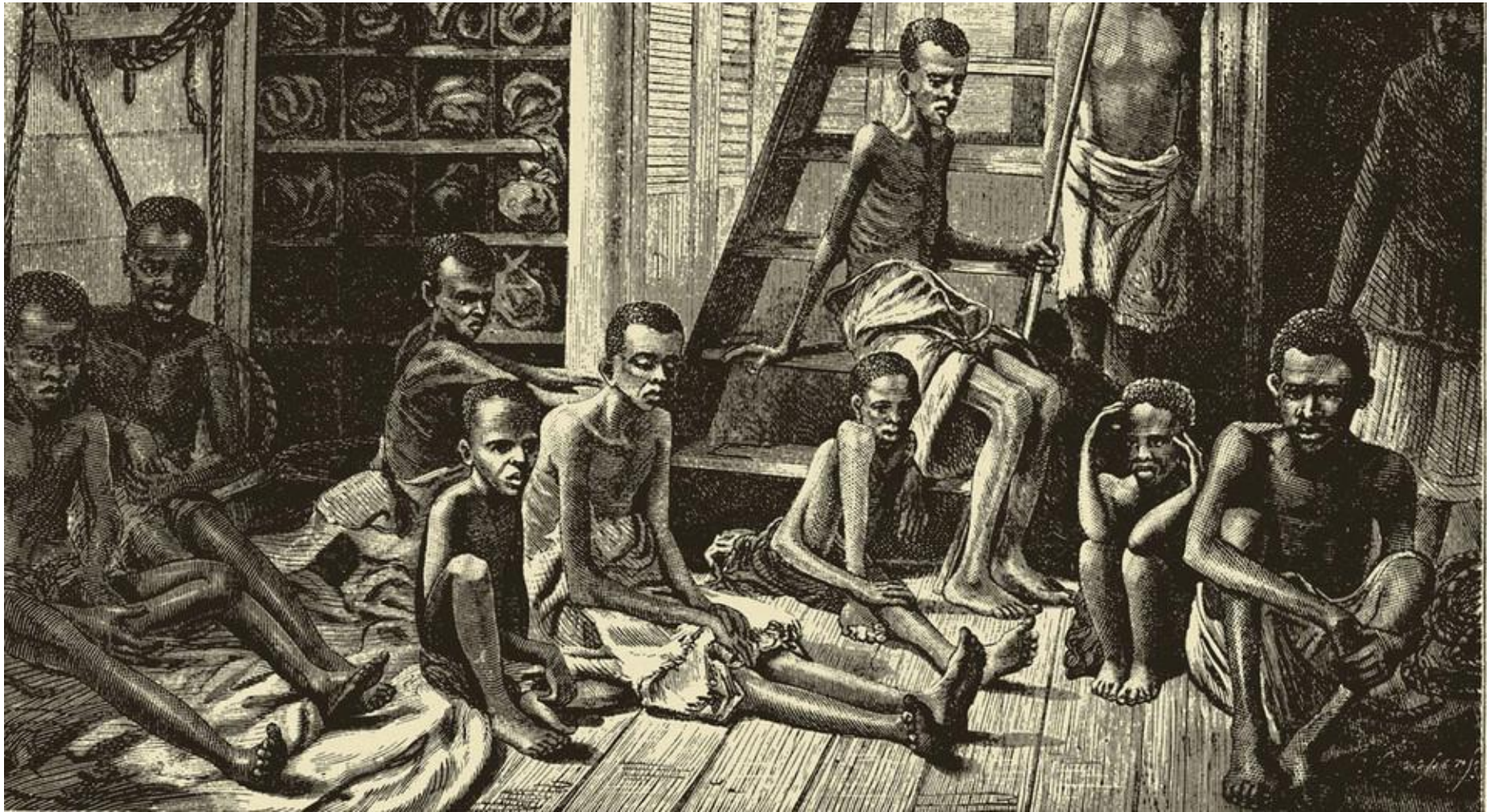
End of the 6th Period
 $86 = 2 \cdot 8 \cdot 18 \cdot 32 \cdot 18 \cdot 8$

End of the 7th Period
 $118 = 2 \cdot 8 \cdot 18 \cdot 32 \cdot 18 \cdot 8 \cdot 2$

Final Quota in Red

Period	Group I	Group II	Group III	Group IV	Group V	Group VI	Group VII	Group VIII	Group IX	Group X	Group XI	Group XII	Group XIII	Group XIV	Group XV	Group XVI	Group XVII	Group XVIII
1	1 H																	
2	2 He	3 Li	4 Be	5 B	6 C	7 N	8 O	9 F	10 Ne									
3	10 Ne	11 Na	12 Mg	13 Al	14 Si	15 P	16 S	17 Cl	18 Ar									
4	18 Ar	19 K	20 Ca	21 Sc	22 Ti	23 V	24 Cr	25 Mn	26 Fe	27 Co	28 Ni	29 Cu	30 Zn	31 Ga	32 Ge	33 As	34 Se	35 Br
5	36 Kr	37 Rb	38 Sr	39 Y	40 Zr	41 Nb	42 Mo	43 Tc	44 Ru	45 Rh	46 Pd	47 Ag	48 Cd	49 In	50 Sn	51 Sb	52 Te	53 I
6	54 Xe	55 Cs	56 Ba	57 La	58 Ce	59 Pr	60 Nd	61 Pm	62 Sm	63 Eu	64 Gd	65 Tb	66 Dy	67 Ho	68 Er	69 Tm	70 Yb	71 Lu
7	86 Rn	87 Fr	88 Ra	89 Ac	90 Th	91 Pa	92 U	93 Np	94 Pu	95 Am	96 Cm	97 Bk	98 Cf	99 Es	100 Fm	101 Md	102 No	103 Lw

Drawings of slave ship Brookes by British abolitionist William Elford. 1788

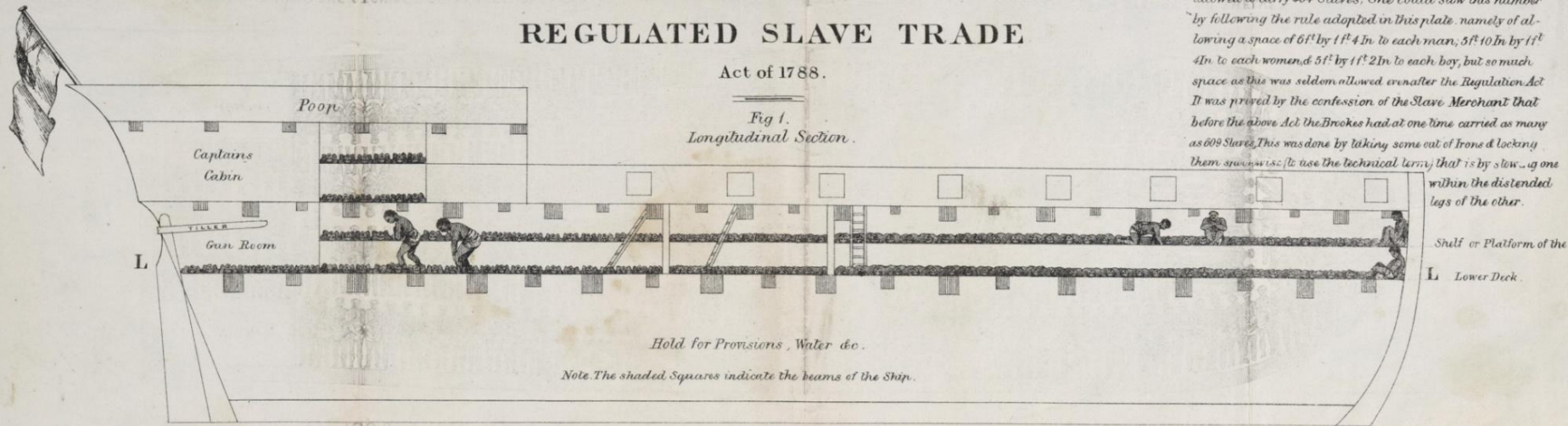


Drawings of slave ship *Brookes* by British abolitionist William Elford. 1788

STOWAGE OF THE BRITISH SLAVE SHIP "BROOKES" UNDER THE REGULATED SLAVE TRADE

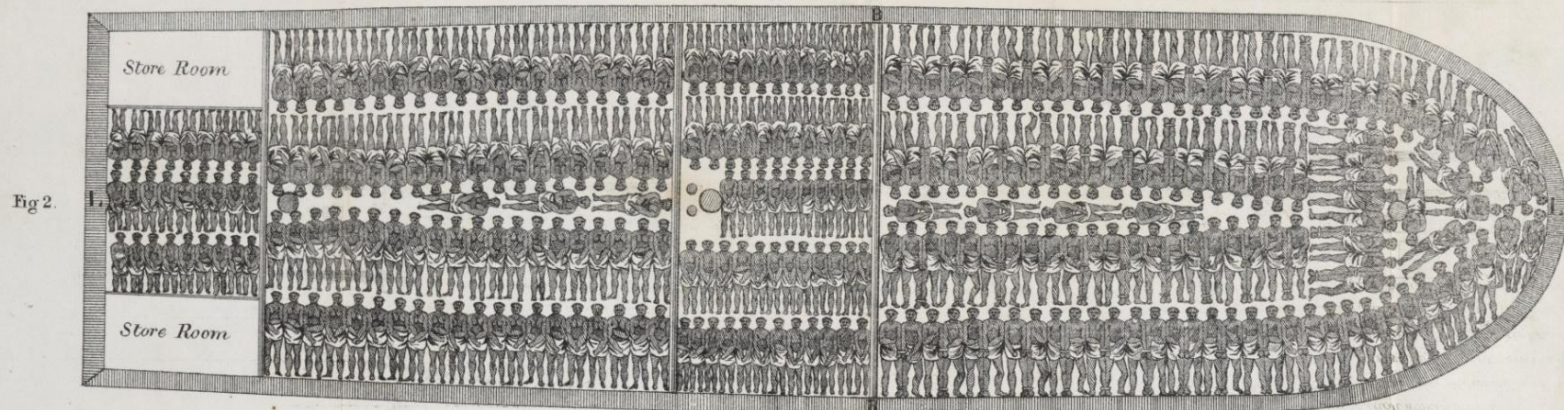
Act of 1788.

Fig 1.
Longitudinal Section.



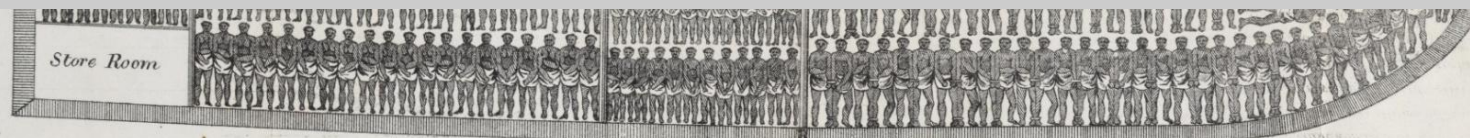
PLAN OF LOWER DECK WITH THE STOWAGE OF 292 SLAVES

130 OF THESE BEING STOWED UNDER THE SHELVES AS SHEWN IN FIGURE B & FIGURE 3.



PLAN SHEWING THE STOWAGE OF 130 ADDITIONAL SLAVES ROUND THE WINGS OR SIDES OF THE LOWER DECK BY MEANS OF PLATFORMS OR SHELVES

Drawings of slave ship *Brookes* by British abolitionist William Elford. 1788



PLAN SHEWING THE STOWAGE OF 130 ADDITIONAL SLAVES ROUND THE WINGS OR SIDES OF THE LOWER DECK BY MEANS OF PLATFORMS OR SHELVES (IN THE MANNER OF GALLERIES IN A CHURCH) THE SLAVES STOWED ON THE SHELVES AND BELOW THEM HAVE ONLY A HEIGHT OF 2 FEET 7 INCHES BETWEEN THE BEAMS: AND FAR LESS UNDER THE BEAMS. See Fig 1.

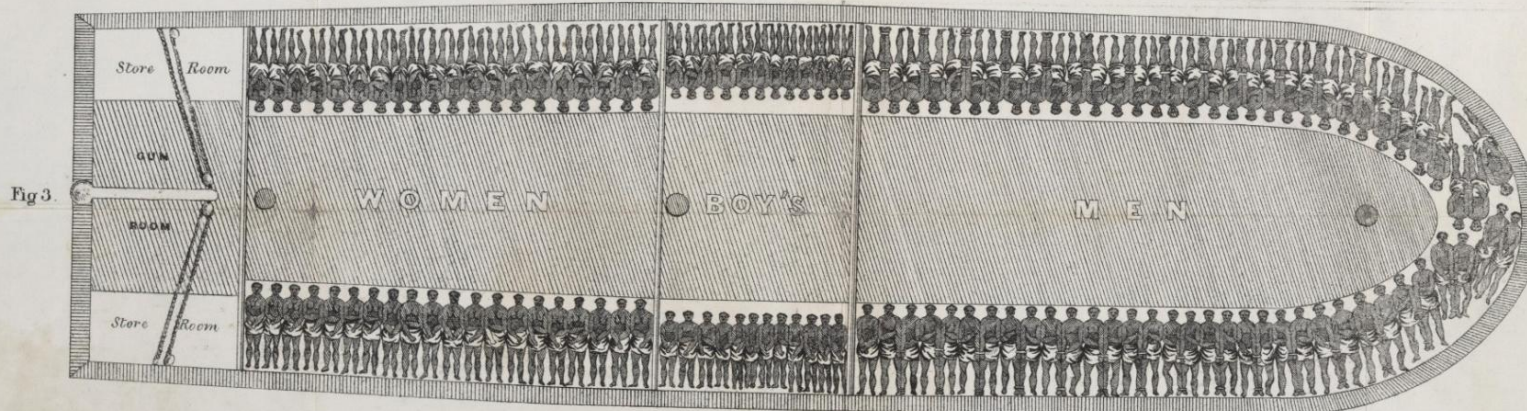


Fig 4.
Cross Section
at the Poop.

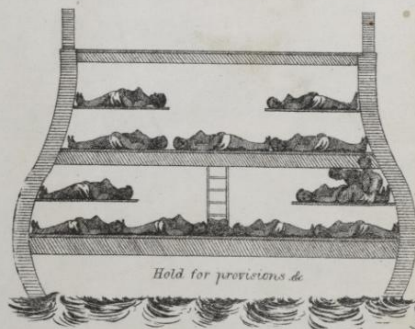


Fig 5.
Cross Section
amidships

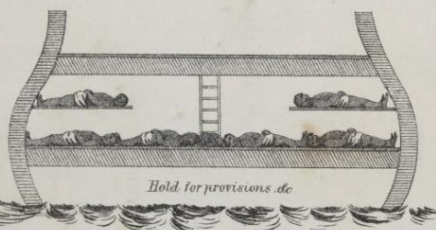


Fig 6.
Lower tier of Slaves under the Poop.

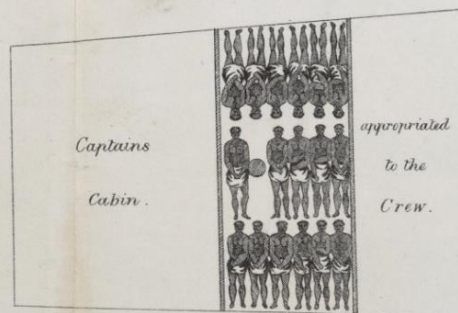
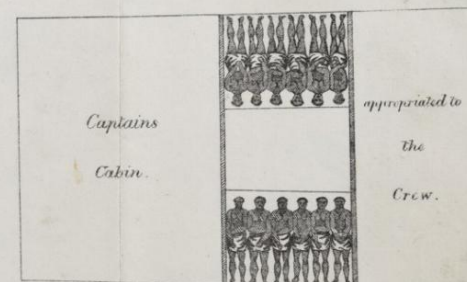


Fig 7.
Shelf tier of Slaves under the Poop.



Scale of Feet
0 5 10 20 30

What is visualization?

Clear thinking made visible – Edward Tufte

It is not about designing graphics. It is all about designing information
– Richard Saul Wurman

Vision can no longer be employed simply to support verbal and conceptual meanings: Its potential as a cognitive power in its own right must be exploited – Kepes

Visualization is really about external cognition, that is, how resources outside the mind can be used to boost the cognitive capabilities of the mind – Stuart Card

Why visualize?

To convey information through visual representations in order to:

- Map/record
- Clarify/explain
- Instruct/communicate
- Discover
- Inspire/tell a story

// PROGRAMM

MITTWOCH 21.04.2004

	12:00	13:00	14:00	15:00	16:00	17:00
LAGERHALLE						
KUNSTHALLE DOMINIKANERKIRCHE						

DONNERSTAG 22.04.2004

	12:00	13:00	14:00	15:00	16:00	17:00
LAGERHALLE	BRAVE NEW WORLD 12:00 // 58 min	ELSEWHERE 13:30 // Le Gout du Koumiz // 66 min		MEMORIES ARE MADE OF THIS 14:30 // 84 min	JONAS AT THE OCEAN 16:00 // Peter Scappell / Xiam / 2002 // 85 min	
ARTHOUSE 4					DIE PERLE IN DER KACKE 16:30 // Dirk BSE // 80 min	EWIG 17:30 // -KURZ-
ARTHOUSE 5					NIWATORI WA HADASHI DA 16:30 // Arima Marisaki / Japan 2002 // 114 min // 0m11	ARAG 17:30 // / Farb
HAUS DER JUGEND (HDJ)	STUDENT FORUM: MEDIA ACADEMIES / PART I 12:00 TAMKE Kyoto Christa Sommerer (A/JP) 13:00 Academy of Fine Arts, Prague Anastasia Morozkina (S/RU)			STUDENT FORUM: MEDIA ACADEMIES / PART II 15:00 TFG Offenbach Bernard Fapo (D) 16:00 AKT Bielefeld RBE Synthesen (NL) 17:00 KEM Köln Karin Peters (D)		

FREITAG 23.04.2004

	12:00	13:00	14:00	15:00	16:00	17:00
LAGERHALLE	SUICIDE 12:00 // American Travelogue // 70 min	OFU) BALANCE 13:30 // 60 min		RETROSPECTIVE: C. MACLAINE 14:30 // 10 min // 60 min	VISIONS OF DELIGHT 16:00 // 58 min	
ARTHOUSE 4					EWIGE SCHÖNHEIT 16:30 // Marcel Schuster // 90 min // mit Vor- film: -SchülerInnen-	DIE P 17:30 //
ARTHOUSE 5					VAMPIRE HUNTER 16:30 // Yoshiki Kawajiri / Japan 2000 // Artime // 105 min // 35mm // Deutsche Fassung	NIWA 17:30 //
HAUS DER JUGEND (HDJ)	TRANSMITTER PART I 12:00 Dr. Karja Kewisch, Uli München // Irrkometen 13:00 Ken Fiegold (USA) // Doc-Art 14:00 Tim Filders/Chaos Computer Club (D) // -Einkaufslicht-			TRANSMITTER PART II 15:30 Fadelman (D) // Transmitter, Transgender, RTUs 16:30 Marie Rutz Gell (E) // The Media Artist 17:30		
DGB-GEBÄUDE						

DAMAGE

SOLUTION

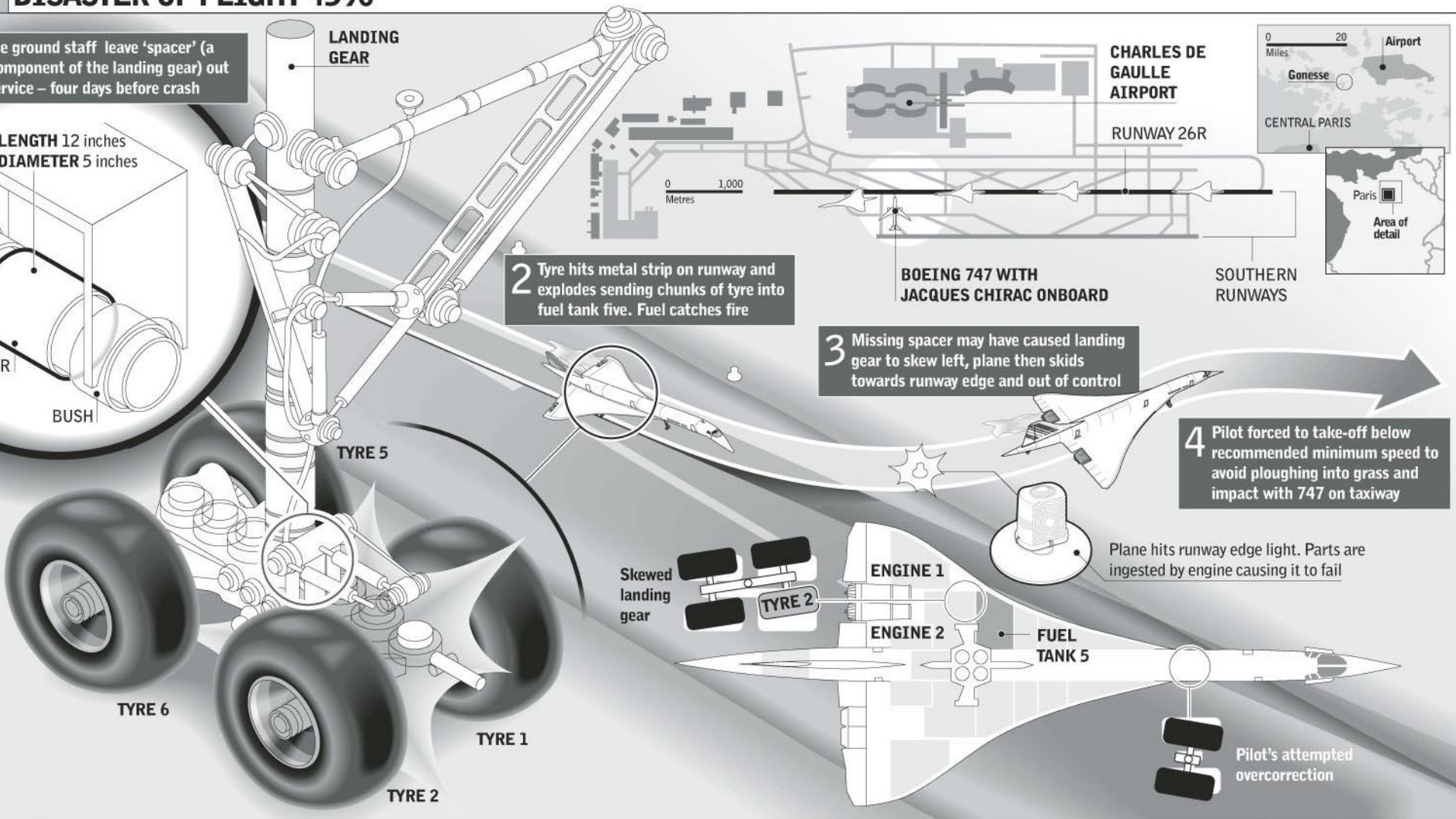
ARE WE REALLY SAFE? ...

pi River levels:

Note: Figures constructed by treatment using MARS, also using Rutherford's et al. model. MARS takes model MARS model as "True" and the cross head model (shown below) is constructed as the model's cross. Therefore, are constructed.

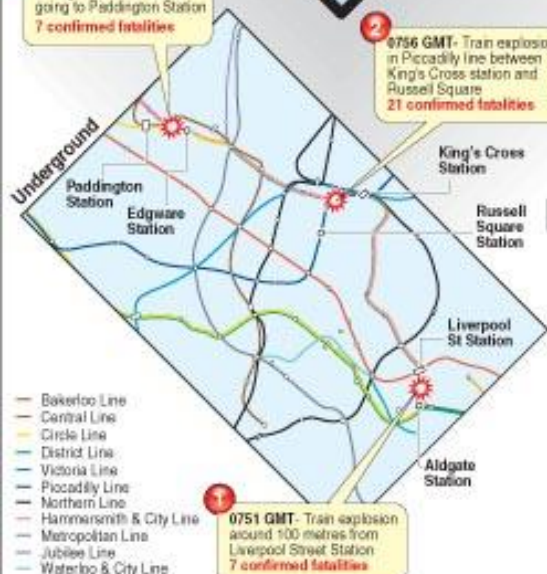
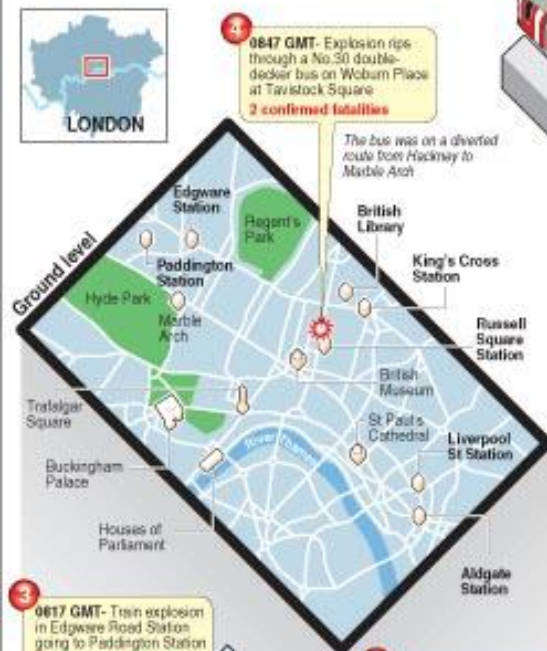
Source: 1992 *London World Business Yearbook*. Some figures in brackets, staff in round.

DISASTER OF FLIGHT 4590

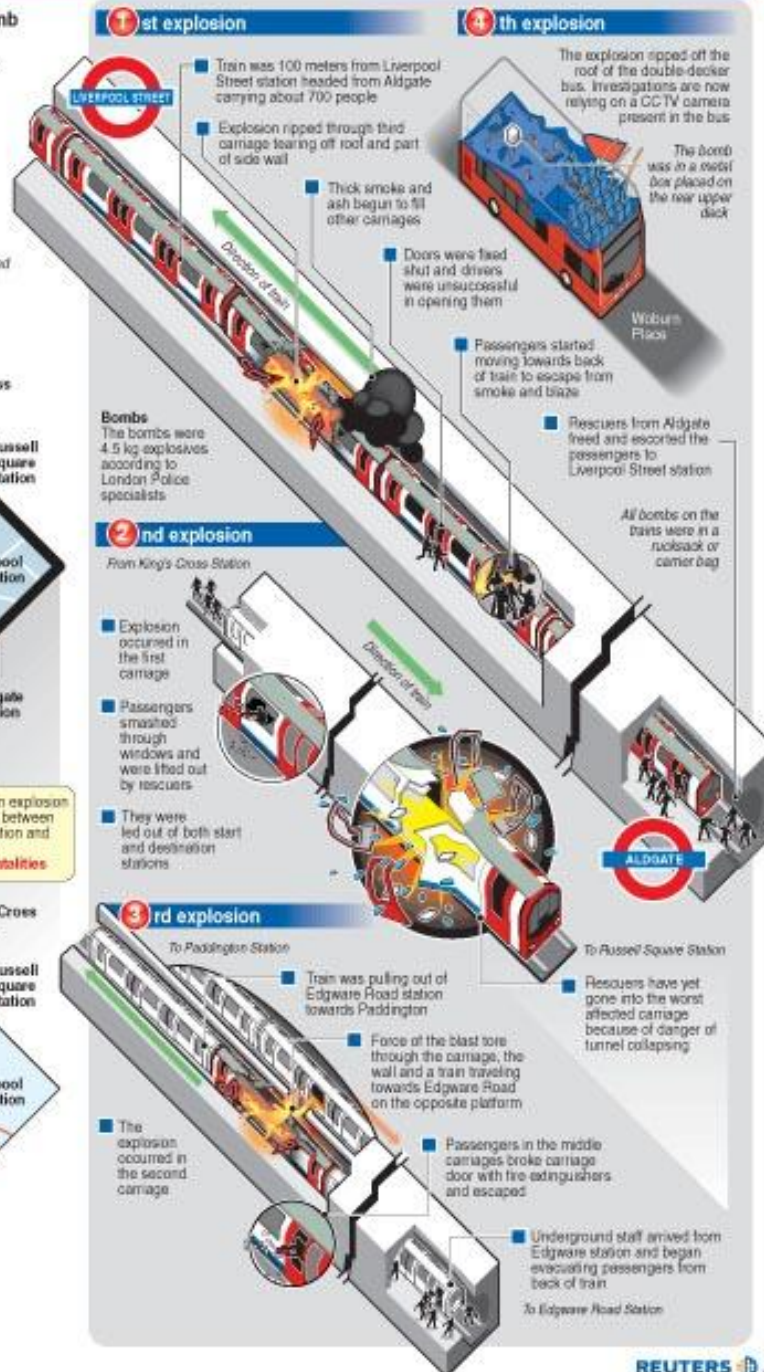


LONDON BLASTS

More than 50 people were killed and 700 injured in bomb attacks on a bus and several underground trains in London on Thursday. Officials believe the blasts bore hallmarks of an al-Qaeda terrorist attack



Source: London Metropolitan Police, Guardian Unlimited



REUTERS

SUPERintrigante

Perguntas instigantes, respostas surpreendentes

BOCA EM AÇÃO COMO É O TRÁFICO NA FAVELA?

Os pontos de tráfico de drogas, conhecidos como "bocas", operam como empresas, escondidos em favelas e bairros pobres das grandes cidades. Os criminosos se organizam em uma hierarquia preocupada em garantir duas coisas: o abastecimento constante de cocaína, maconha e outros entorpecentes e o sistema de proteção contra a polícia ou quadrilhas rivais.

Para garantir a eficiência do negócio, são contratados diversos funcionários. O esquema de segurança e a acirrada disputa entre traficantes põem em risco a vida de compradores e moradores da favela. "Até chegar à boca, o usuário tem que andar na favela. Ele é avaliado e nem percebe. Se os seguranças pensarem que ele é um policial disfarçado, atiram", diz o delegado Carlos Roberto Alves de Andrade, da Delegacia de Repressão ao Crime Organizado do Departamento de Narcóticos de São Paulo. **5**

por: Cássio da Silva

CRÍME ORGANIZADO VÁRIOS FUNCIONÁRIOS ESTÃO ENVOLVIDOS NO ESQUEMA DE TRÁFICO

AVIOZINHOS

Os garotos que levam a droga da boca para os clientes são mais comuns no Rio de Janeiro. Em São Paulo, onde as favelas são planas, a distância entre a boca e o consumidor é pequena e o serviço deles nem sempre é necessário.

ALTO ESCALÃO

Traficantes de maior hierarquia ficam posicionados sobre lajes e barracos, onde podem se proteger melhor e atirar em caso de tentativa de invasão. Carregam fuzis, ideais para combates a longa distância.

A BOCA

Geralmente fica perto de riachos, esgotos ou barrancos, para dificultar a chegada da polícia. Em uma mesma favela, podem existir várias bocas e nem toda a droga fica aqui. Barracos conhecidos como "paiol" são usados para armazenamento de grandes quantidades e da munição da quadrilha.

GERENTE DA BOCA

É responsável pela chegada da droga e pela contratação do pessoal. É ele que comanda toda a operação dentro da favela e, por isso, é sempre alguém de muita confiança do dono da boca.

SEGURANÇAS

A função deles é proteger os arredores da boca da polícia e de traficantes rivais. Eles usam armas próprias para combater a curta distância.

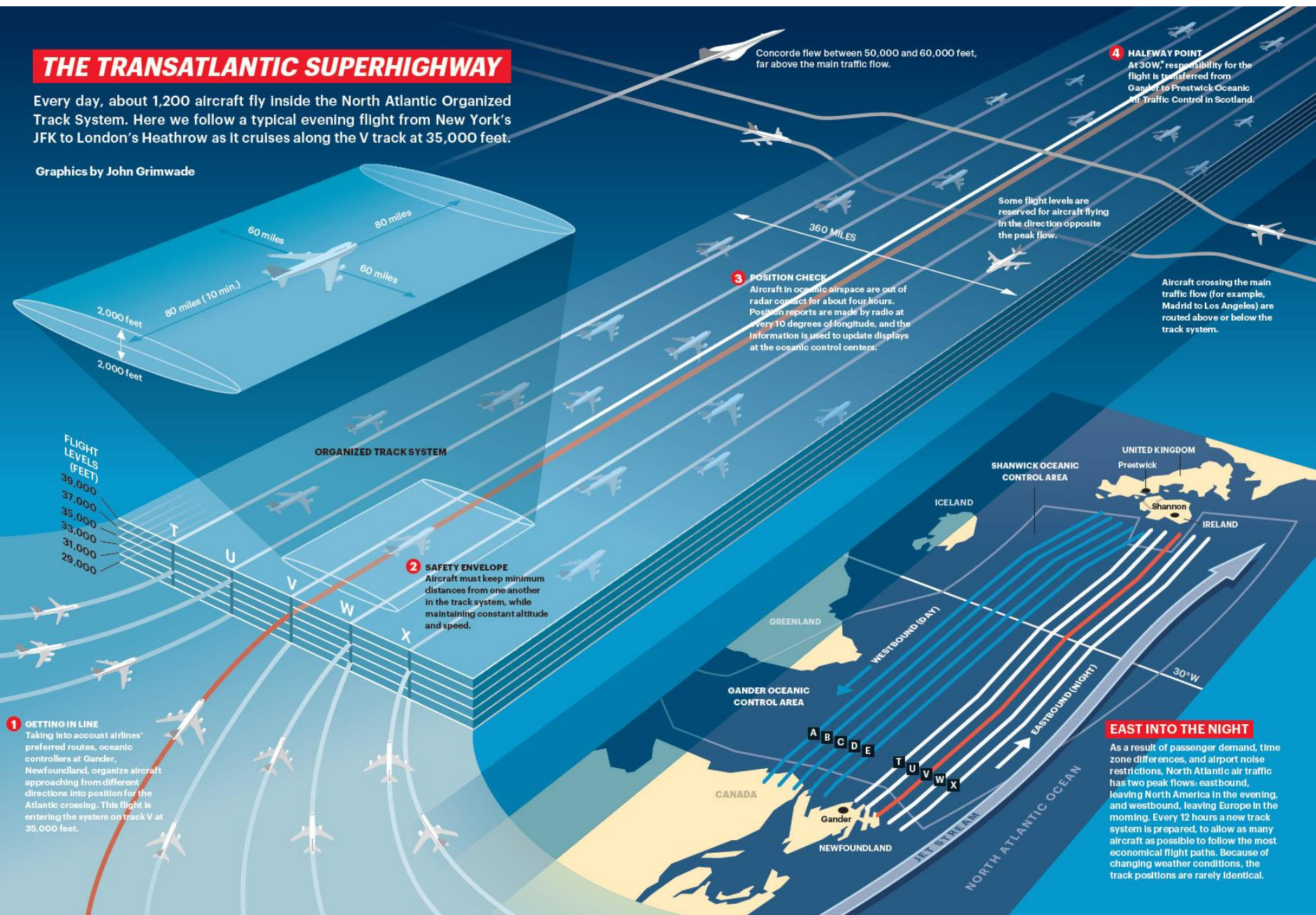
ENQUANTO ISSO...

O dono da "boca" não lida diretamente com a venda da droga. Ele comanda o tráfico de um barraco ou casa afastada, por meio dos gerentes. Bocas bem-sucedidas podem transformar traficantes em homens ricos e bem de vida.

THE TRANSATLANTIC SUPERHIGHWAY

Every day, about 1,200 aircraft fly inside the North Atlantic Organized Track System. Here we follow a typical evening flight from New York's JFK to London's Heathrow as it cruises along the V track at 35,000 feet.

Graphics by John Grimwade



1 GETTING IN LINE
Taking into account airlines' preferred routes, oceanic controllers at Gander, Newfoundland, organize aircraft approaching from different directions into position for the Atlantic crossing. This flight is entering the system on track V at 35,000 feet.

2 SAFETY ENVELOPE
Aircraft must keep minimum distances from one another in the track system, while maintaining constant altitude and speed.

3 POSITION CHECK
Aircraft in oceanic airspace are out of radar contact for about four hours. Position reports are made by radio at very 10 degrees of longitude, and the information is used to update displays at the oceanic control centers.

Concorde flew between 50,000 and 60,000 feet, far above the main traffic flow.

4 HALFWAY POINT
At 30°W, responsibility for the flight is transferred from Gander to Prestwick Oceanic Air Traffic Control in Scotland.

Some flight levels are reserved for aircraft flying in the direction opposite the peak flow.

Aircraft crossing the main traffic flow (for example, Madrid to Los Angeles) are routed above or below the track system.

EAST INTO THE NIGHT

As a result of passenger demand, time zone differences, and airport noise restrictions, North Atlantic air traffic has two peak flows: eastbound, leaving North America in the evening, and westbound, leaving Europe in the morning. Every 12 hours a new track system is prepared, to allow as many aircraft as possible to follow the most economical flight paths. Because of changing weather conditions, the track positions are rarely identical.

**BREITLING
EMERGENCY
WATCH**

**SECONDARY
ANTENNA**
Pulling out the
auxiliary antenna
extends the
transmitter's
range without
drawing more
power.

CRYSTAL
The transmitter
and watch works
are separate.
Increasing the
likelihood that
even if the watch
is damaged in a
crash, the
transmitter may
function.

CROWN

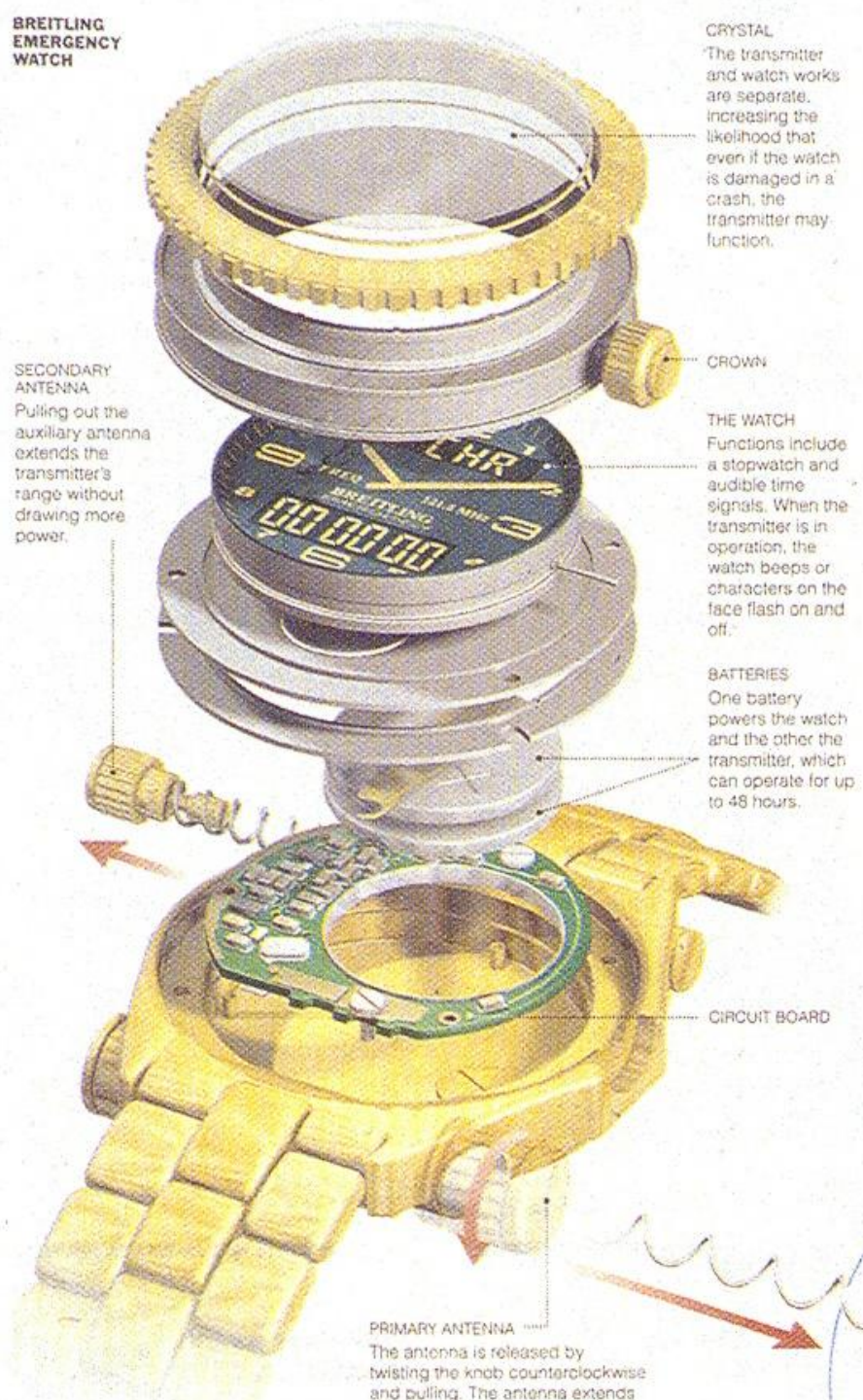
THE WATCH
Functions include
a stopwatch and
audible time
signals. When the
transmitter is in
operation, the
watch beeps or
characters on the
face flash on and
off.

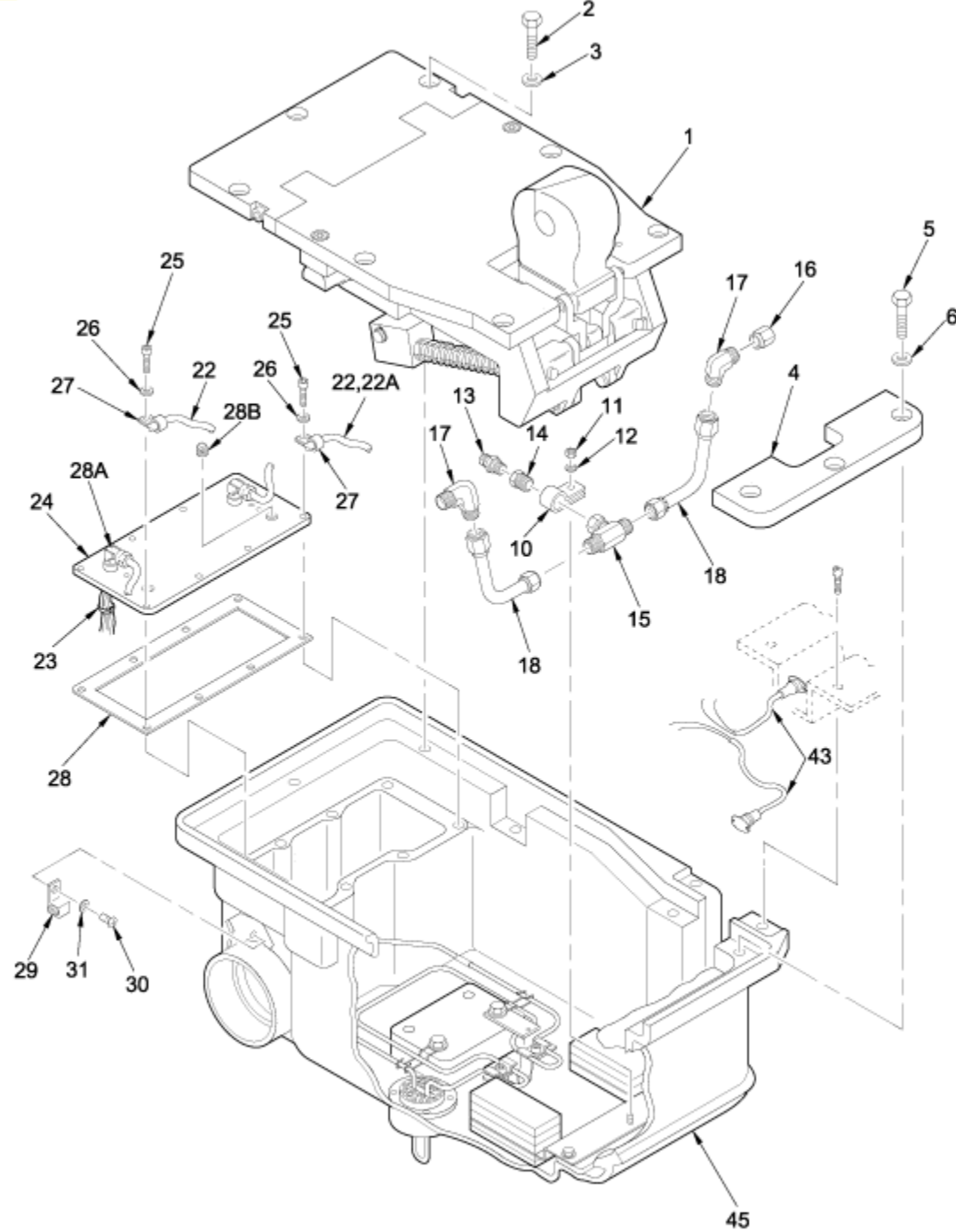
BATTERIES
One battery
powers the watch
and the other the
transmitter, which
can operate for up
to 48 hours.

CIRCUIT BOARD

PRIMARY ANTENNA

The antenna is released by
twisting the knob counterclockwise
and pulling. The antenna extends





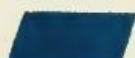
Oliver Byrne's *Elements of Euclid*. 1847

44

BOOK I. PROP. XLIII. THEOR.



THE complements

 and  of the parallelograms which are about the diagonal of a parallelogram are equal.



Q. E. D.

BOOK I. PROP. XLIV. PROB.

45



O a given straight line (—) to apply a parallelogram equal to a given tri-

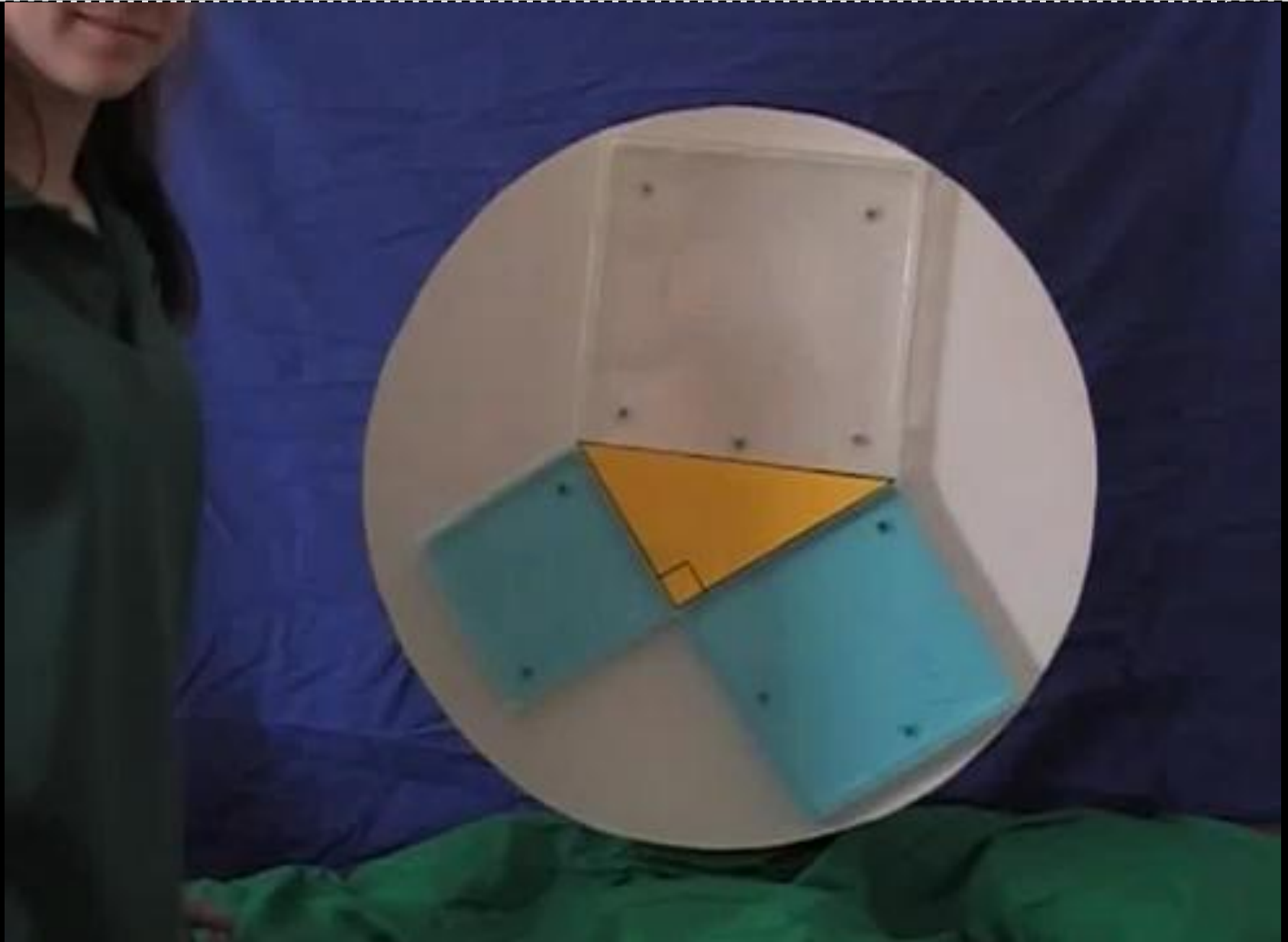
angle (), and having an angle equal to a given rectilinear angle ().



Make  =  with  =  (pr. 42.)
and having one of its sides ----- conterminous with and in continuation of ————.
Produce ———— till it meets ———— || -----
draw ----- produce it till it meets ----- continued;
draw ----- || ----- meeting ----- produced, and produce -----.



A tangible visualization of Pythagorean theorem



space required to transport 60 people



car



bus



bicycle

Nadeem Haidary's visualization of caloric consumption per capita



PIZZAEXPRESS

1 car space = 10 bicycles





Hungry Planet: What The World Eats – Peter Menzel
<https://time.com/8515/what-the-world-eats-hungry-planet/>



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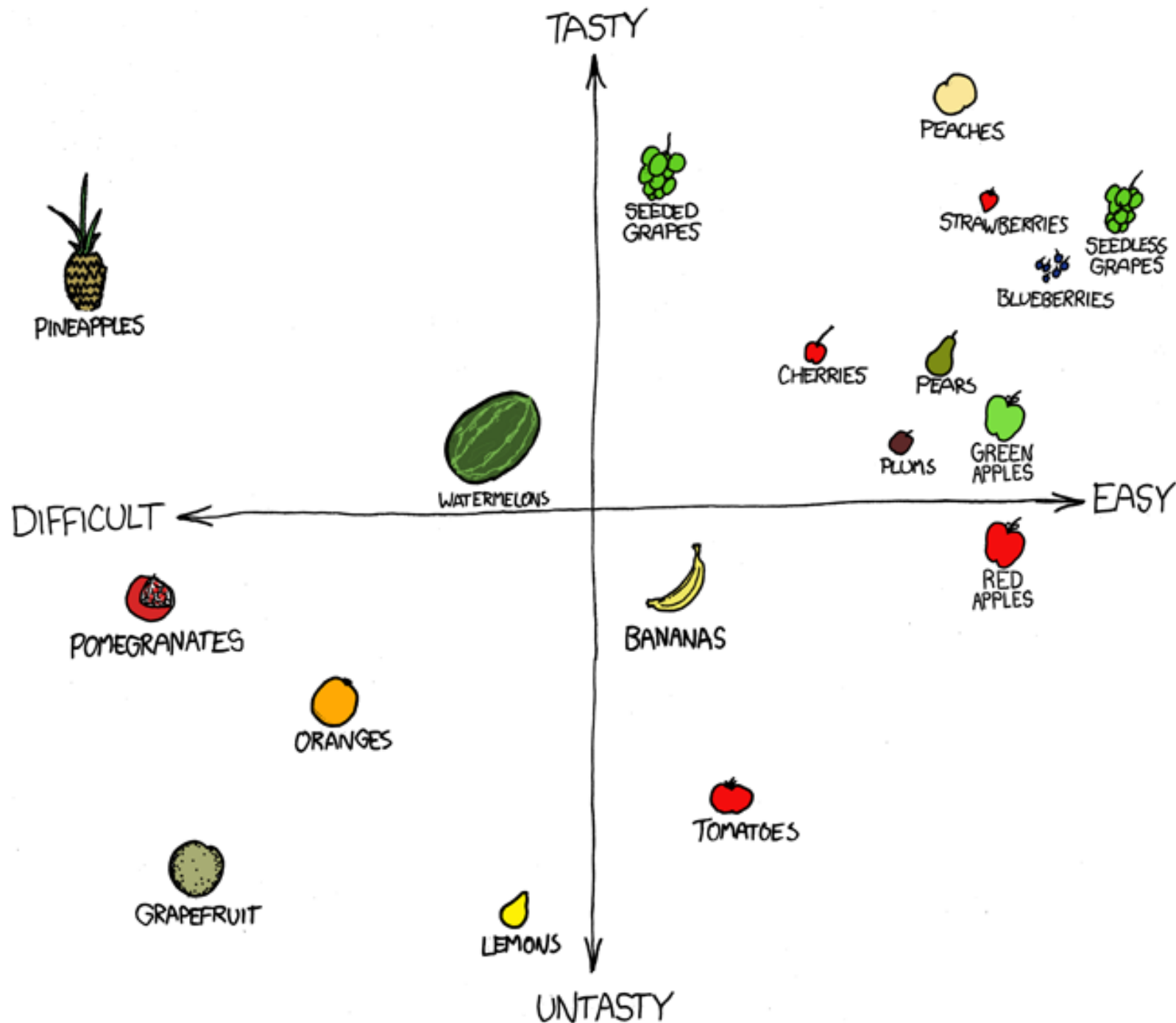
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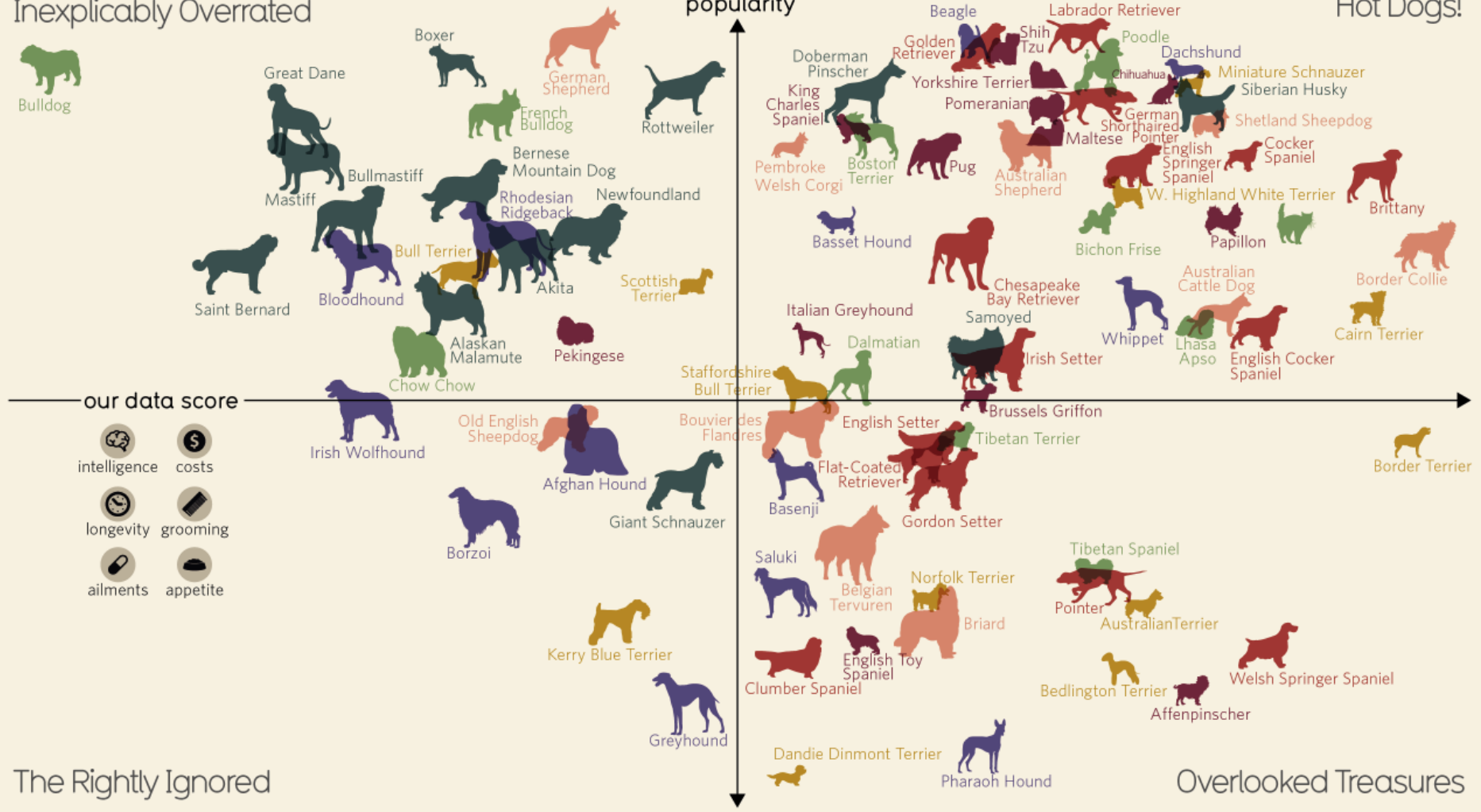
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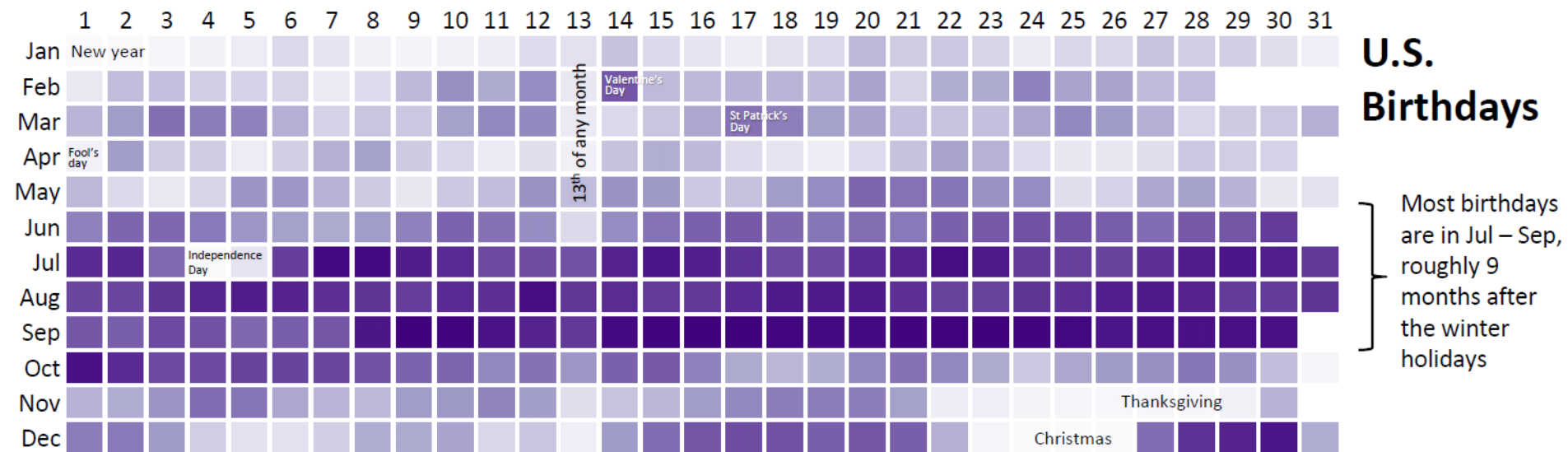


Best in Show: The Ultimate Data Dog

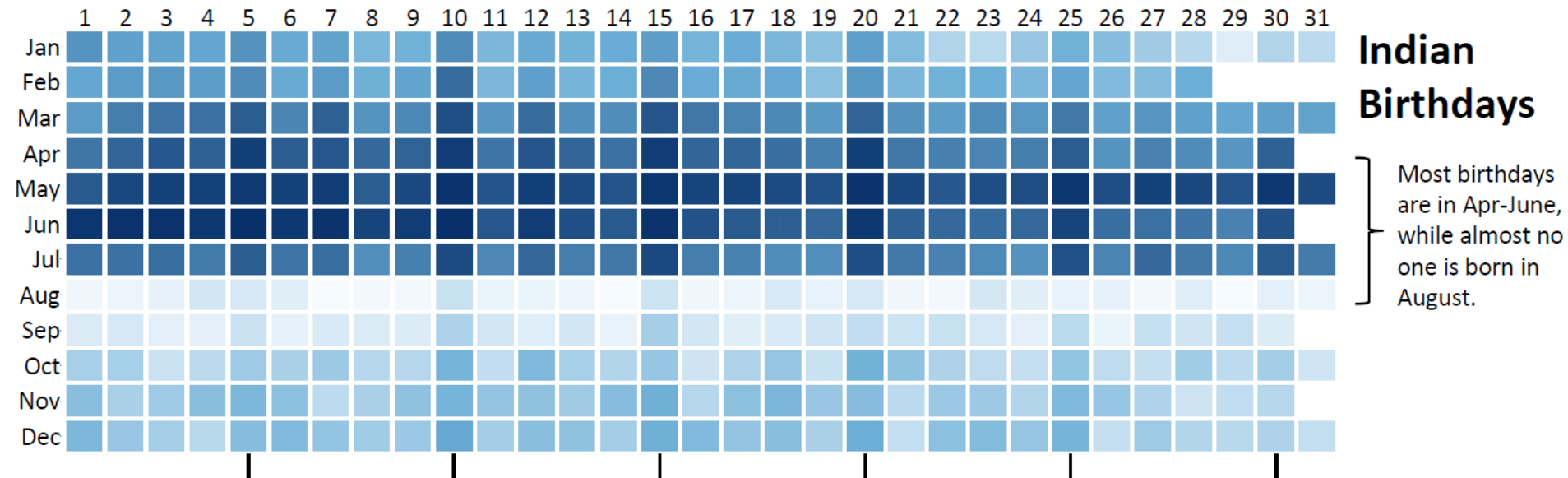
Inexplicably Overrated



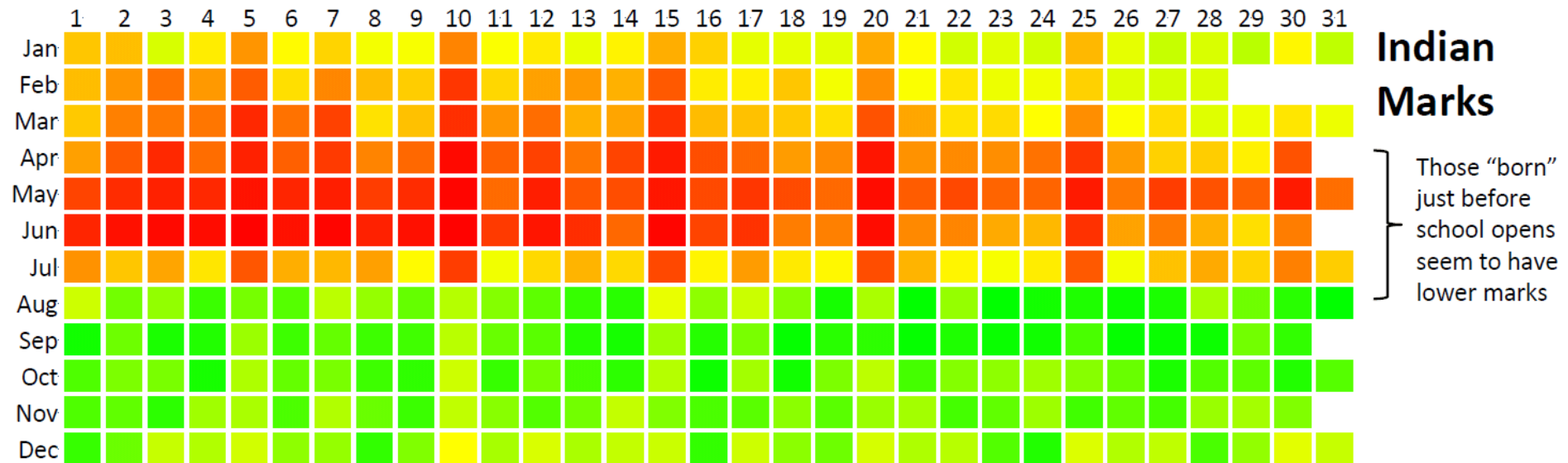
Birthdays in the U.S and in India, Gramener



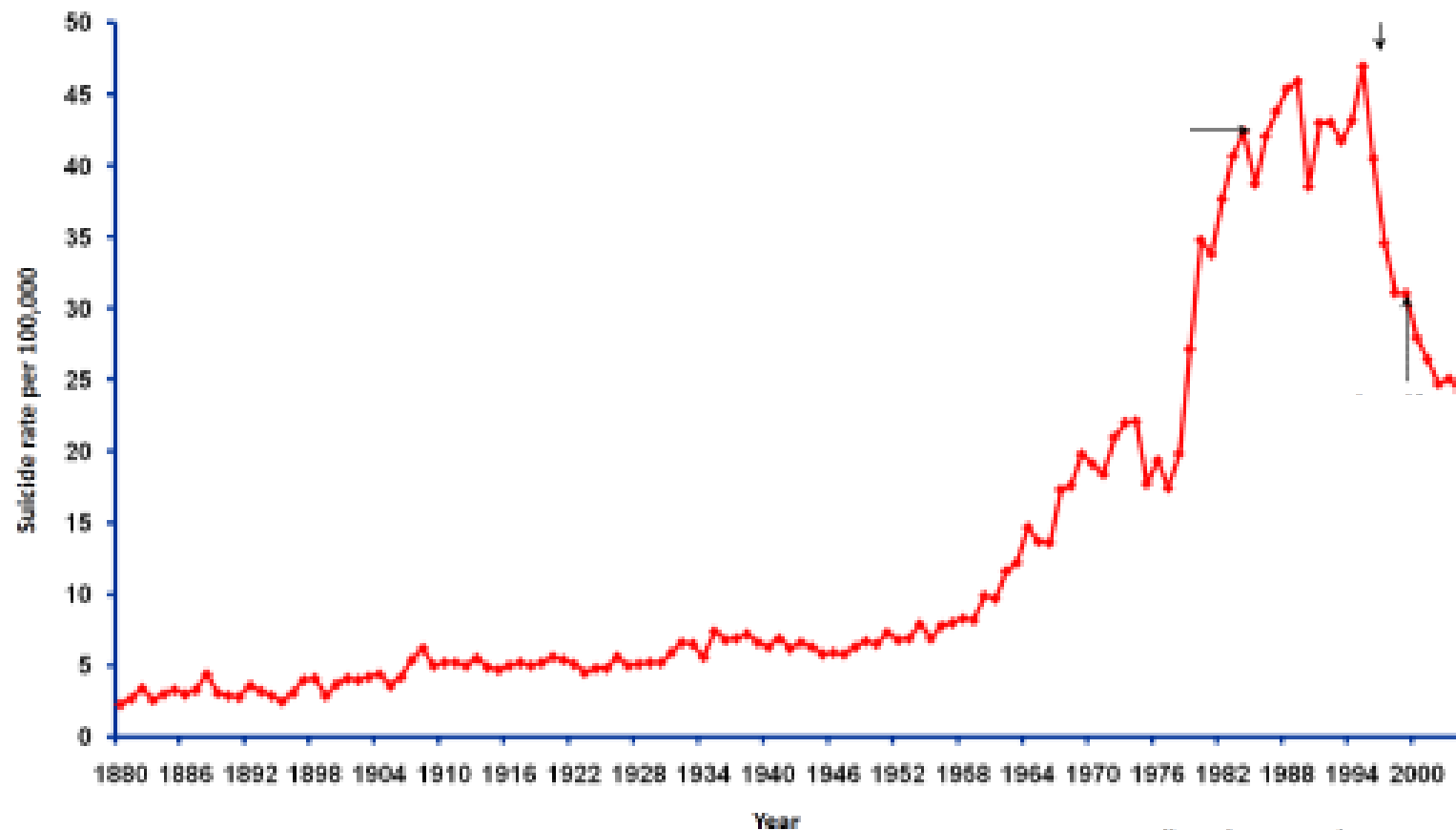
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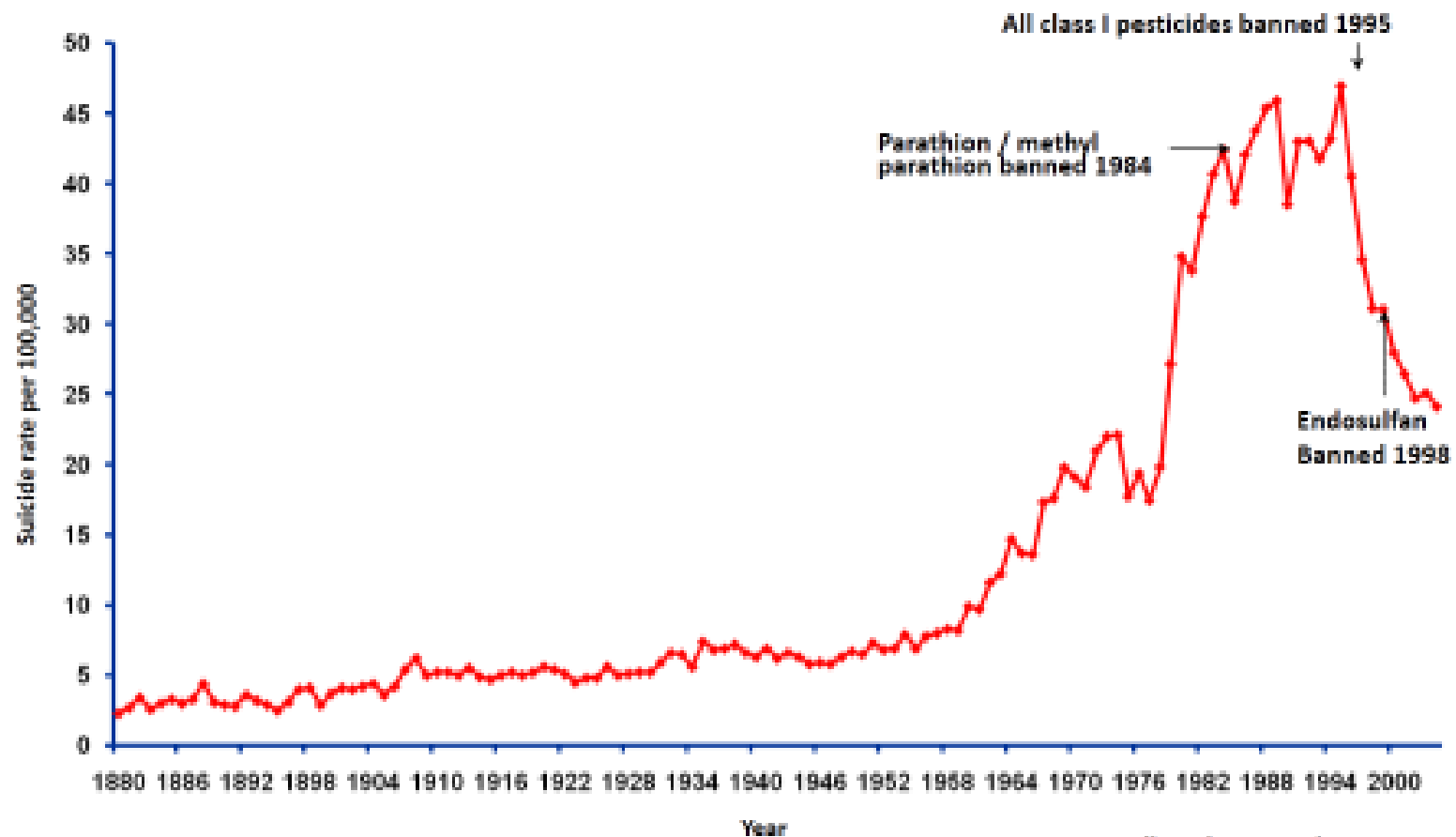


Suicide rates in Sri Lanka 1880-2005



Gunnell *et al*, *Int J Epid* 2007

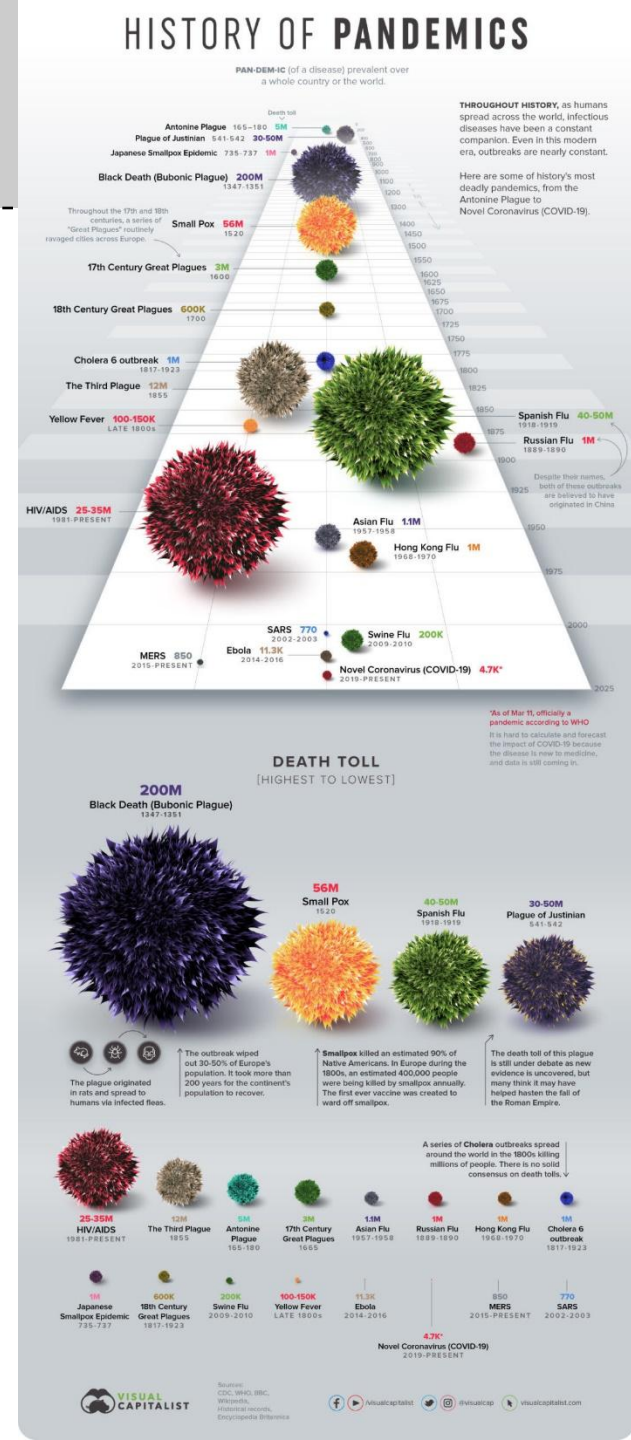
Suicide rates in Sri Lanka 1880-2005



Gunnell *et al*, *Int J Epid* 2007

Assignment 1

<https://www.visualcapitalist.com/history-of-pandemics-deadliest/>



Assignment 1

- Comparison of length, area, or volume?
- Fuzzy edges
- Colour coding?
- Shape coding?
- Semantic coding?
- *Ponzo illusion* (linear perspective - unless blobs are created inside some strictly observed three-dimensioned model of space)
- Annotations?
- Should it be relative to time? How does 1 million in 6 years compare with half a million in 6 weeks?
- Should it be adjusted for era? (5 million deaths from 165 to 180 AD would be something like 180 million today)
- 7 of them are caused by bacteria, not viruses

